

disse demo

research showcase II

Tao 🤠



- 3rd year CS PhD working on **HCI** w/ **Lydia Chilton**
- Research on human-AI interaction design & creativity support tools
- TA-ed UI design, Designing for GenAI, Advanced Web Design Studio
- From Shenzhen, China, Undergrad @ Cornell
 - information science & communication
 - graduated in 2022, so i'm not that old lol, i also listen to olivia rodrigo & many kpop hehe
- Talk to me about...
 - anything :)
 - NYC concerts (i went to 5 concerts last summer and helped hold one in central park)
 - food (especially the all you can eat ones)
 - free trips (i went for free trips almost every 2 months)
 - PhD (maybe!)

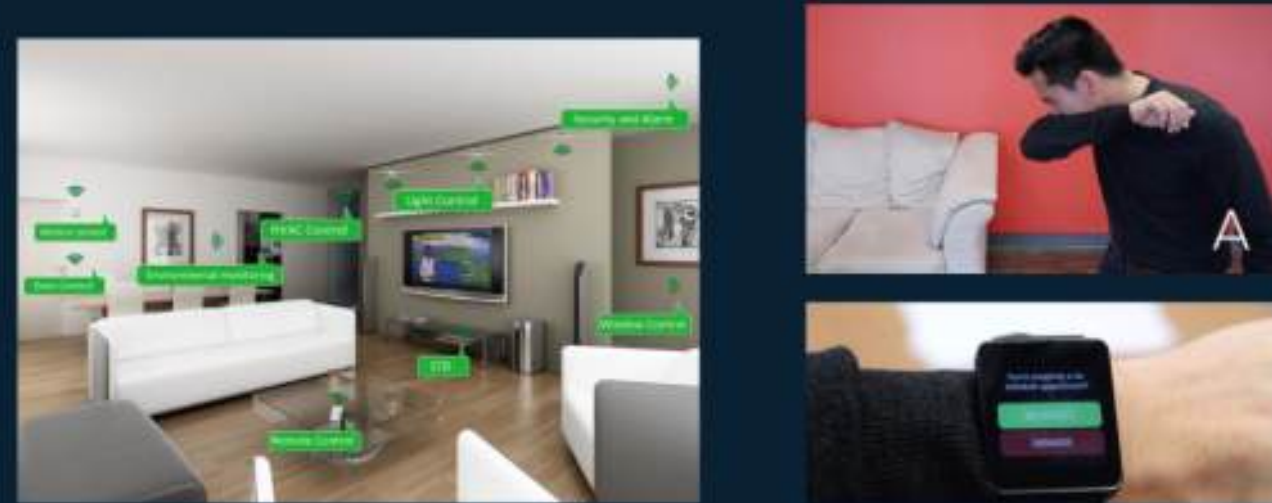
Human-Computer Interaction (HCI)

Cross-disciplinary field that studies how computers should be designed

- What computers are used for
- How people interact with computers
- How either of these two things should change in the future

Human-Computer Interaction (HCI)

2. Ubiquitous Computing and Mobile Health



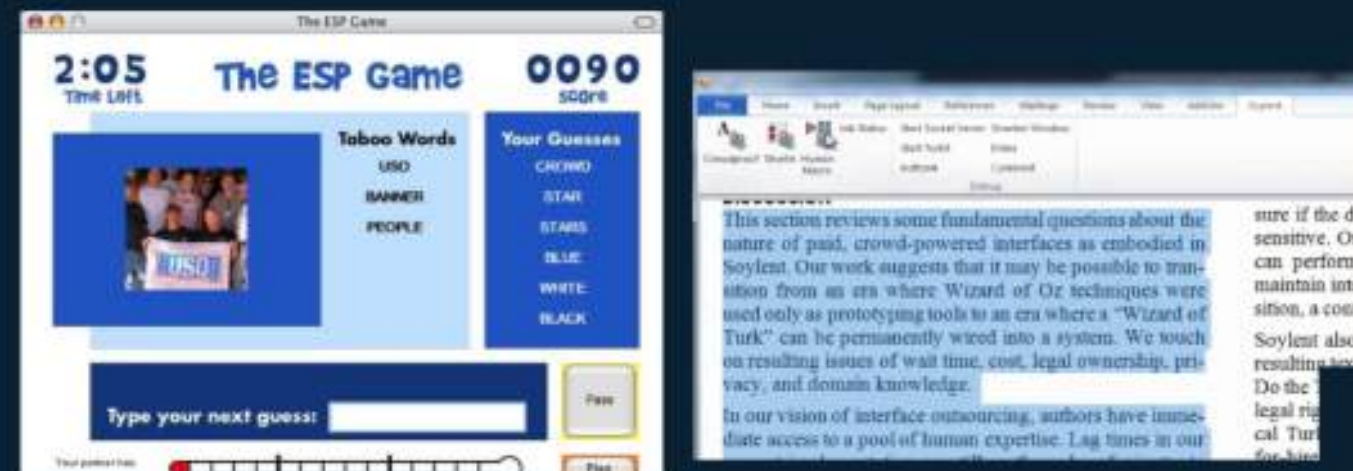
4. Tangible Computing and New Interaction Styles



5. Mixed Reality and Wearable Computing



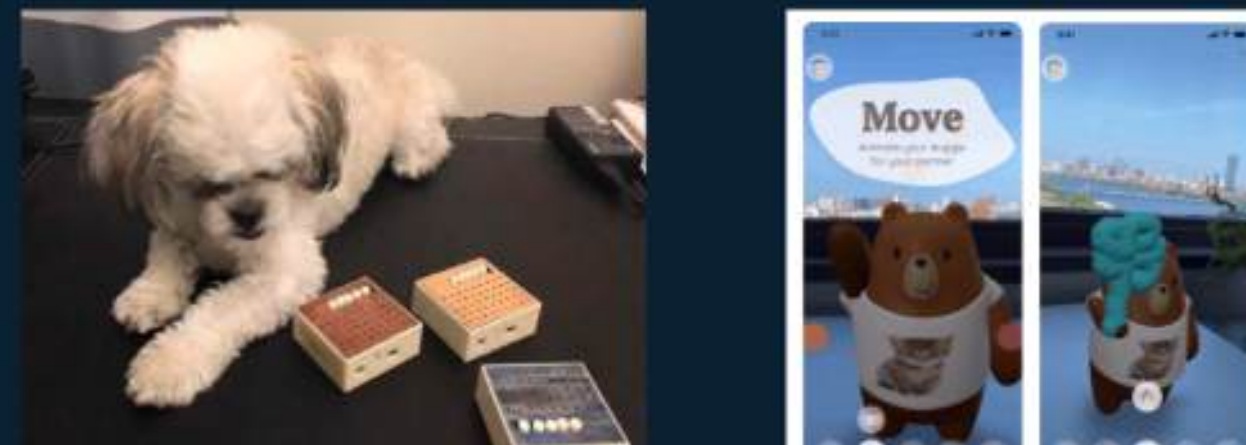
6. Collective Intelligence



11. Computational Interaction and Generative AI



7. Social Computing



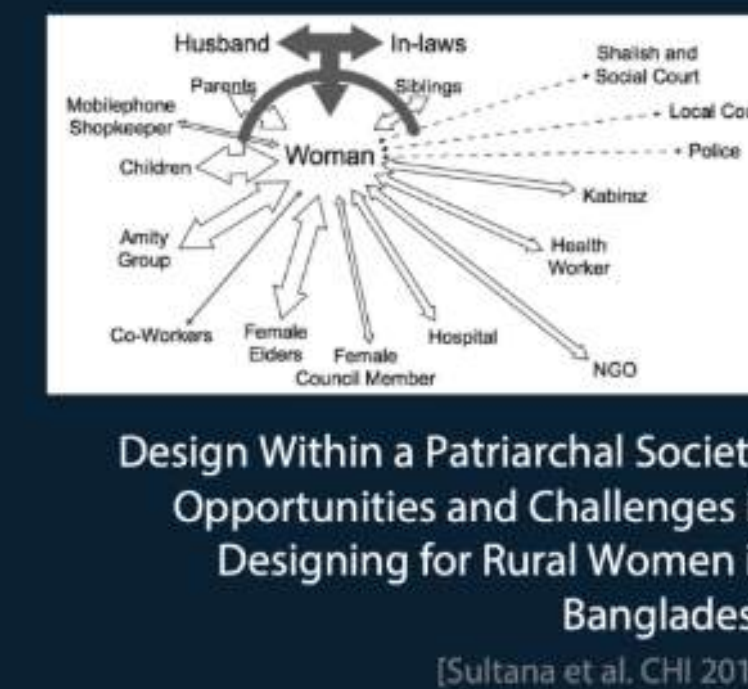
12. Sensemaking and Creativity Support



8. Accessibility

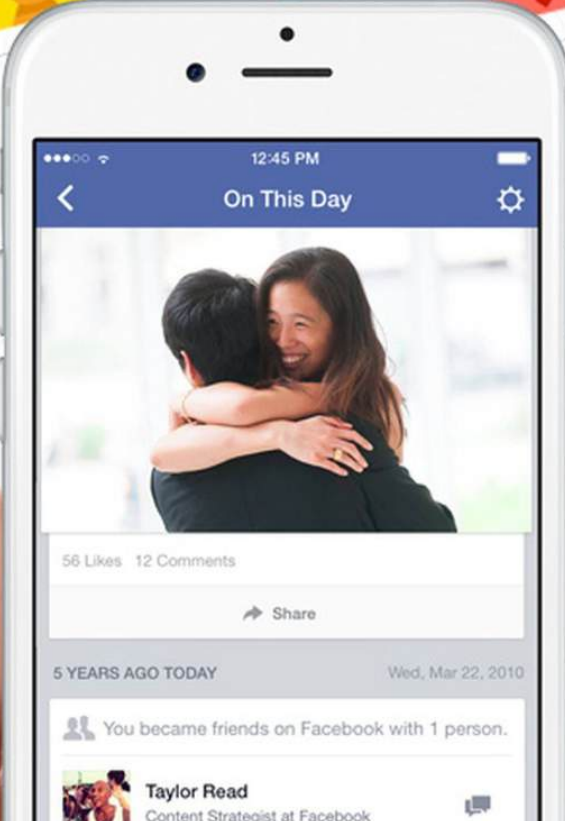


13. HCI and Society



Something fun (from HCI)

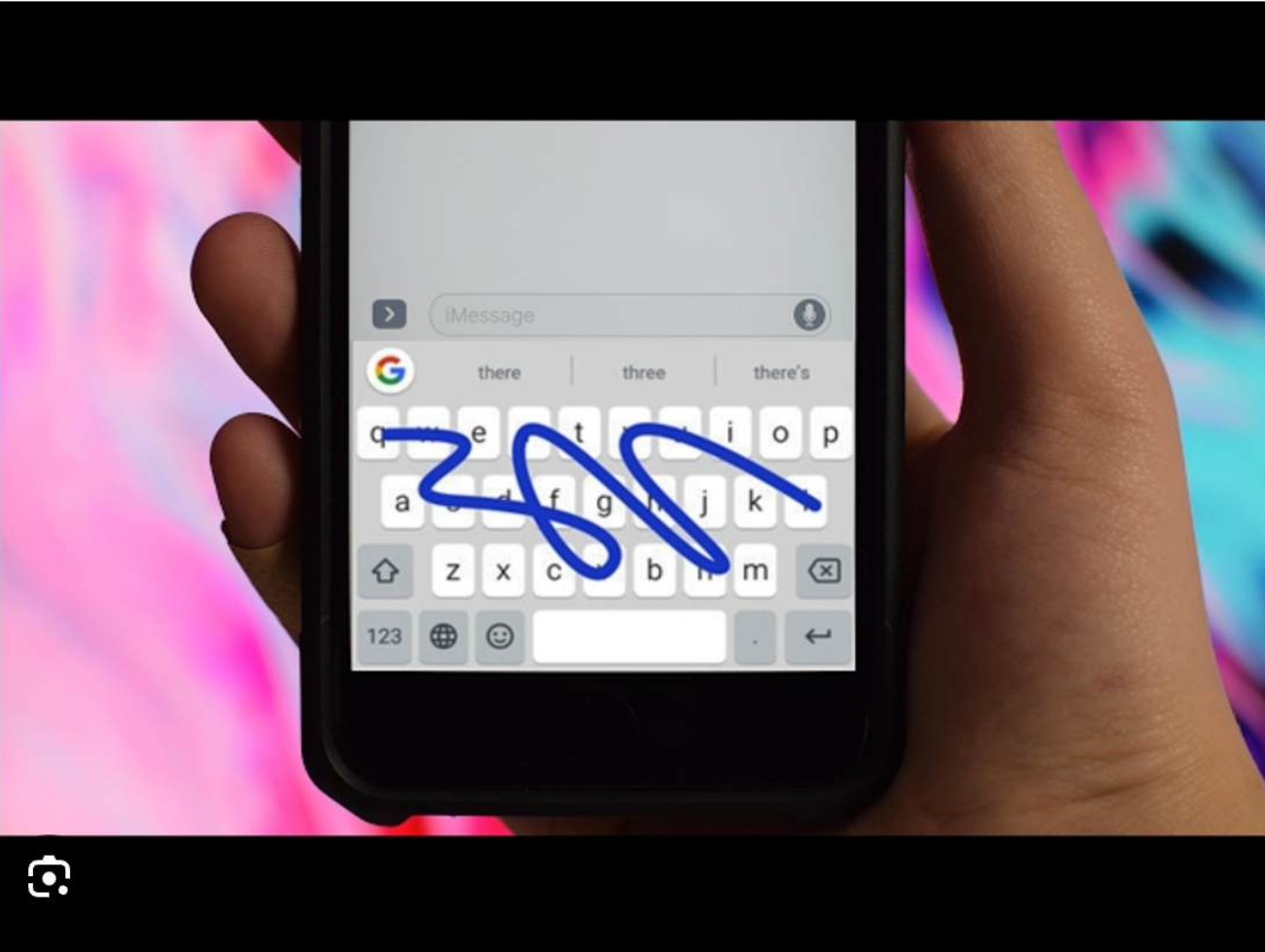
The Verge



Facebook officially launches nostalgia-inducing 'On This Day' feature - The Verge

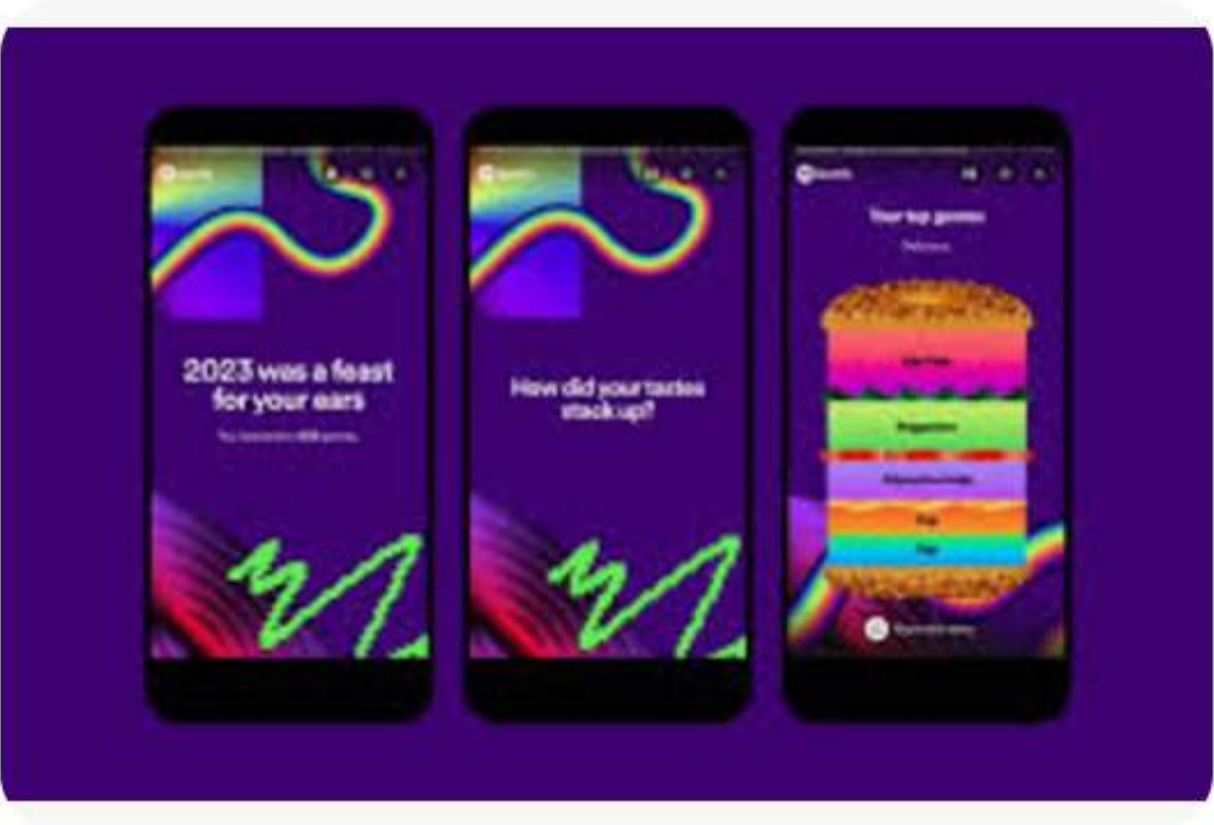
Visit >

YouTube

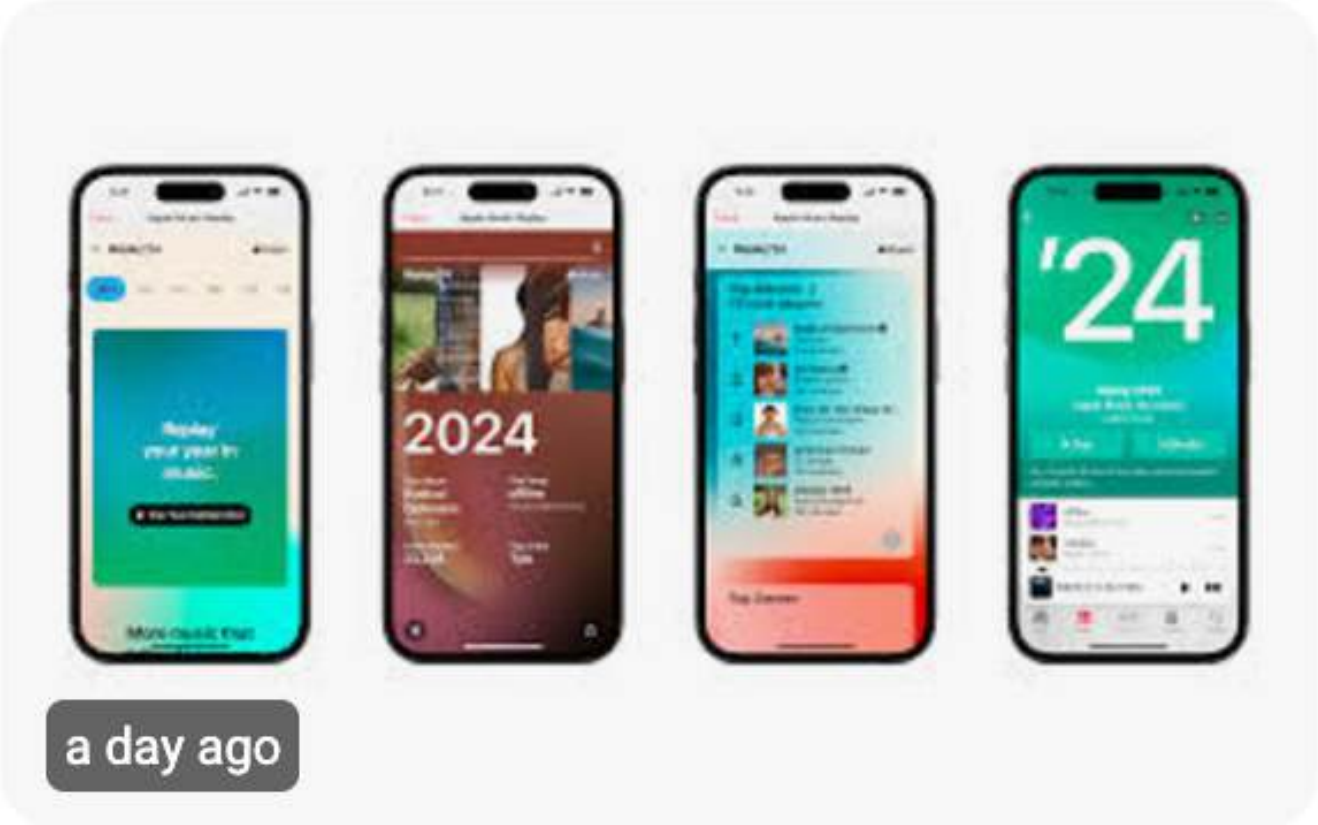


Swipe Typing For Your iPhone! - YouTube

Visit >



Florida Today
Spotify Wrapped 2024 released. Here'...



TechCrunch
Apple Music's 2024 Replay rolls out to ...



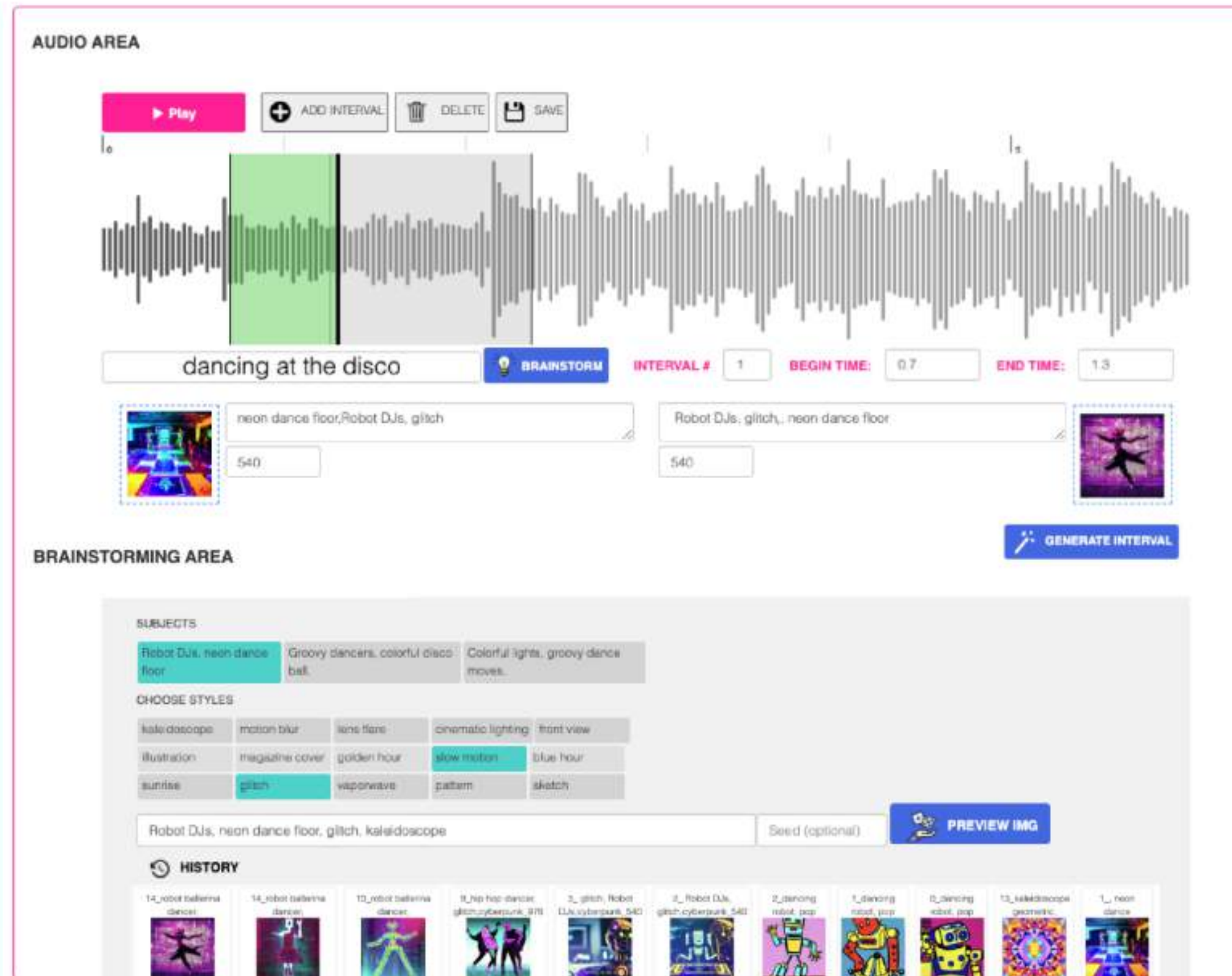
My research: creativity support and human-AI interaction

- How do creative professionals use AI?
 - (ie artists, writers, journalists(?), content creators in general)
- What are their experiences, challenges, and perception?

***I build creativity support tools to
assist creative professionals***

writing tool, design tool, etc.

My research: creativity support and human-AI interaction



VIDEO AREA



EMPIRE STATE OF MIND - Alicia Keys





Tweertorial Hook Incubator

Step 1: Input Topic & Generate Everyday Examples

[tweet_topic]:

prompt:

language models

Please list 5 everyday examples of [tweet_topic]. Each example should be under 10 words, related to a general audience, and not require technical knowledge.

START

Here is a Quick Hook Generation:

Ever wonder how your phone predicts what you're about to type next? 🤖 That's thanks to language models! Think of it as your device's mini fortune teller, guessing your next word based on what you've typed so far. But how can it predict your personal vibs? Look out for our upcoming blog to dive into the mystery! 🕵️ #LanguageModelsAreAwesome

Here are 5 everyday examples:

- Auto-correcting typos on your smartphone. ❤️
- Predictive texting while writing messages. 📩 I Copied to Next Step
- Voice recognition like Siri or Alexa. 🗣️
- Search engine suggestions. 🔍
- Email spam filters. 📧

REGENERATE

Step 2: Generate Common Experiences

[everyday_example]:

prompt:

Predictive texts and plot suggestions for creative writing

Please list 5 common experiences with using [everyday_example]. Each experience should be under 10 words, related to a general audience, and not require technical knowledge.

GENERATE COMMON EXPERIENCES

Here are 5 common experiences:

- Unexpectedly humorous or nonsensical suggestions. 🤖 I Copied to Next Step
- Autocorrect mistakes changing intended meaning. 📧
- Helpful in faster typing and spelling correction. 📝
- Stunting creative expression with standard phrases. 🗣️
- Learning to use unusual words or phrases less often. 📖

REGENERATE

Step 3: Generate User Scenarios

[common_experience]:

prompt:

Unexpectedly humorous or nonsensical suggestions.

Please list 12 user scenarios someone encounters with [common_experience] while using [tweet_topic]. Each scenario should be under 15 words, from a first-person perspective, related to a general audience, and not require technical knowledge.

GENERATE USER SCENARIOS

Here are 5 user scenarios:

- While writing an email, it suggested I end with "Rise like a dinosaur!" 🦖
- It replaced "Good morning" with "Party with unicorns." 🦄
- It suggested to rephrase "Will you marry me?" with "Can you be my penguin?" 🐧
- Using it as a recipe guide, it added "bake sauce for 5 mins" in cooking steps. 🍳
- It replaced "meeting at 3pm" with "mammoth cave tour at sunset." 🦕
- When typing "Guardians of", it completed as "Guardians of the gummy bears." 🍬 I Copied to
- While typing a complaint letter, it suggested "Let's have a pizza party!" 🍕
- Asked about the weather, it replied me with "Cloudy chance of flying saucers." 🛸
- Documenting meeting minutes, it recommended "Meeting was an alien conspiracy." 🛸
- It suggested citing "Report released by Invisible College of Fairy Engineers" in my assignment. 🧚
- Writing a heartfelt poem, it ended with "And so, the broccoli whisped." 🥦
- Asked for the synonym of "happy", it suggested "cheerful cheerleaders." 🧑

REGENERATE

Step 4: Generate Personal Anecdotes

[user_scenario]:

prompt:

When typing "Guardians of", it completed as "Guardians of the gummy bears"

Please write 3 short personal anecdotes about the topic, [tweet_topic], for the user scenario of [user_scenario] for creative writing. Each anecdote should be under 30 words, from a creative writer's perspective, related to a general audience, and not require technical knowledge. Make sure to mention that the writer actually incorporated the funny suggestion for their future plot.

GENERATE PERSONAL ANECDOTES

Here are the 3 personal anecdotes:

- While crafting a superhero novel, I typed "Guardians of..." intending to type "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gummy Bears". It sparked an idea for a humorous subplot. ❤️ I Copied to Next Step
- I was creating a mythical saga and paused at "Guardians of...". Hilariously, the language model completed it as "Guardians of the Gummy Bears". I worked it in as a hilarious microcosm. 🗣️
- Absentmindedly, passing "Guardians of..." into a political satire, the language model proposed "Guardians of the Gummy Bears". I laughed out loud and turned it into a whimsical political parody. 🗣️

REGENERATE

Step 5: Generate More Specific Anecdotes

[personal_anecdote]:

prompt:

While crafting a superhero novel, I typed "Guardians of..." intending to type "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gummy Bears". It sparked an idea for a humorous subplot.

Please rewrite 3 versions of this anecdote to make it more specific. Replace all generic terms with concrete and specific instances! [personal_anecdote].

GENERATE SPECIFIC ANECDOTES

Here are 3 specific versions:

- While I was diligently typing up my latest Batman-inspired book, my fingers swiftly moved to type the famous phrase "Guardians of Gotham City". However, my cheeky automated spellcheck unexpectedly suggested "Guardians of the Gumball Machine". This surprise sparked an intriguing concept for a side-splitting subplot to my main plot. 🦇
- Once, during an intense session of composing my Marvel-based masterpiece, I began to write "Guardians of the Cosmic Gem". Calling back to my cherished "Guardians of the Galaxy" characters, but my computer, with its AI language model, hilariously replaced it with "Guardians of the Candy Cane". This hilarious mix-up sparked a brilliant idea for an uproariously funny tangent within my main story. 🍬
- One day, while I was passionately crafting an Avengers-centric narrative, I was all set to reference the iconic "Guardians of the Infinity Stone". Suddenly, my quirky autocorrect amusingly provided the suggestion "Guardians of the Jelly Beans". This humorous curvetal ignited an idea for a comical thread weaving its way through the main narrative. 🍬

REGENERATE

Step 6: Generate Example Hooks

[specific_anecdote]:

prompt:

Last week, while crafting a sci-fi novel, I typed "Guardians of..." intending to type "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gummy Bears". It sparked an idea for a humorous subplot. (I must say WHY NOT MAKE IT SWEET? And with a giggle I wrote a whole chapter about the gummy bear kingdom. 🍬🍬🍬)

Here is a personal anecdote about the topic, [tweet_topic], for the experience of [common_experience]: [specific_anecdote].
Based on this anecdote, please write 3 versions of the opening tweet to introduce the concept of [tweet_topic]. The tweet should have no jargon. It should contain a specific and relatable example. It should have a driving question to spark curiosity and make readers want to read more. Make sure to include a few emojis and all the details.

GENERATE EXAMPLE HOOKS!

Here are 3 example hooks generated by GPT:

- Ever typed "Guardians of" and expected "the Galaxy" but got "the Gummy Bears" instead? 🤖 What if I told you, this was no random suggestion but the workings of a brilliant technology called "Language Models"? Get your interest piqued? Then follow me into this enlightening thread! 🧵👉
- As I penned down an epic sci-fi novel last week, my writing assistant threw a curveball by suggesting "Guardians of the Gummy Bears"! 🍬 Can you believe it? It made me chuckle, as well as ponder, so, what's generating these unexpected yet fun suggestions? Venture with me into the realm of "Language Models". / s/
- Picture this: You're mindlessly keyboarding away, and out pops a suggestion to silly, it makes your day! Like when I was drafting my sci-fi saga and my device suggests "Guardians of the Gummy Bears". 🍬 Inspired by these funny suggestions? Let's unravel the magic of "Language Models" together. s/

REGENERATE

Step 7: Write a Final Hook

Ever typed "Guardians of" and expected "the Galaxy" but got "the Gummy Bears" suggestion? 🤖🍬 I just got sugary space heroes featured in my new sci-fi! SO CUTE! or what? I told you, this was no random suggestion but a brilliant technology called "Language Models" 🧠 s/

275/280

TWEET



[tweet_topic]

"language models"

Step 1: Input Topic to Generate Everyday Examples

Step 2: Generate Common Experiences

Step 3: Generate User Scenarios

Step 4: Generate Personal Anecdotes

Step 5: Generate More Specific Anecdotes

Step 6: Generate Example Hooks

Step 7: Write a Final Hook



🍬😷 When I fell ill just before my thesis deadline, drafting an email to ask for an extension was.....



101/280

TWEET

Generative AI workflows brings novel & impressive abilities as they chain AI outputs together with human interaction.

main project page

Tweetorial Hook Incubator

Step 1: Everyday Examples

Step 2: Common Experiences

Step 3: User Scenarios

Step 4: Personal Anecdotes

Step 5: Specific Anecdotes

Step 6: Example Hooks

Step 7: Write a Final Hook

ID	Stored value
[tweet_topic]	language models
[everyday_example]	Predictive texts and plot suggestions for creative writing.
[common_experience]	Unexpectedly humorous or nonsensical suggestions.
[user_scenario]	When typing "Guardians of G" completed as "Guardians of the Galaxy".
[personal_anecdote]	While crafting a superhero novel, I typed "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gumby Bears". It sparked an idea for a humorous subplot.
[specific_anecdote]	Last week, while crafting a sci-fi novel, I typed "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gumby Bears". It sparked an idea for a humorous subplot.
[example_hook]	While crafting a superhero novel, I typed "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gumby Bears". It sparked an idea for a humorous subplot.

Data Table (for reference)

[tweet_topic]:

language models

prompt:

Please list 5 everyday examples of [tweet_topic]. Each example should be under 10 words, related to a general audience, and not require technical knowledge.

START

Here is a Quick Hook Generation:

Ever wonder how your phone predicts what you're about to type next? It's thanks to language models! These AI-powered devices guess your next word based on what you've typed so far. But how can it predict your personal vocab? Look out for our upcoming blog to dive into the mystery! #LanguageModels #AI #JustStartTech

Here are 5 everyday examples:

- Auto-correcting typos on your smartphone.
- Predictive texting while writing messages. Copied to Next Step
- Voice recognition like Siri or Alexa.
- Search engine suggestions.
- Email spam filters.

REGENERATE

Notes (♥ & save ideas)

Wow! Sometimes it is so Content Creation, Creativity!!!

Auto-correcting typos on your smartphone
helpful in faster typing and spelling correction.
I recently saved me by having speech-to-text suggest the right words. LOL!

rephrase "Will you marry me?" with "Can you be my pen pal?"
While crafting a superhero novel, I typed "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gumby Bears". It sparked an idea for a humorous subplot.

Writing essays involves unexpected plot... Recently my best suggestion depicted the future as a world of "Guardians of the Galaxy" instead of "Guardians of the Galaxy". My language model amusingly suggested "Guardians of the Gumby Bears". It sparked an idea for a humorous subplot.

Ready to level up your creativity to legendary? 🚀

Step 2: Generate Common Experiences

[everyday_example]:

predictive texts and plot suggestions for creative writing

prompt:

Please list 5 common experiences with using [everyday_example]. Each experience should be under 10 words, related to a general audience, and not require technical knowledge.

GENERATE COMMON EXPERIENCES

Here are 5 common experiences:

- Unexpectedly humorous or nonsensical suggestions. Copied to Next Step
- Autocorrect mistakes changing intended meaning.
- Helpful in faster typing and spelling correction.
- Stimulating creative expression with standard phrases.
- Learning to use unusual words or phrases less often.

REGENERATE

Step 3: Generate User Scenarios

[common_experience]:

Unexpectedly humorous or nonsensical suggestions.

prompt:

Please list 12 user scenarios someone encounters with [common_experience] while using [tweet_topic]. Each scenario should be under 15 words, from a first-person perspective, related to a general audience, and not require technical knowledge.

GENERATE USER SCENARIOS

Here are 5 user scenarios:

- While writing an email, I suggested I end with "Your life is dinosaur."
- It replaced "Good morning" with "Party with unicorns."
- It suggested to rephrase "Will you marry me?" with "Can you be my penpal?"
- Using it as a recipe guide, it added "fence paint for 5 mins" in cooking steps.
- It replaced "meeting at 3pm" with "mammoth cave tour at sunset."

Step 4: Generate Personal Anecdotes

[user_scenario]:

When typing "Guardians of G" completed as "Guardians of the gumby bears"

prompt:

Please write 3 short personal anecdotes about the topic, [tweet_topic], for the user scenario of [user_scenario] for creative writing. Each anecdote should be under 30 words, from a creative writer's perspective, related to a general audience, and not require technical knowledge. Make sure to mention that the writer actually incorporates the funny suggestion for their future plot.

GENERATE PERSONAL ANECDOTES

Here are the 3 personal anecdotes:

- While crafting a superhero novel, I typed "Guardians of G" intending to type "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gumby Bears". It sparked an idea for a humorous subplot. Copied to Next Step
- I was creating a mythical saga and paused at "Guardians of G". hilariously, the language model completed it as "Guardians of the Gumby Bears". I worked it in as a fantastical micronoun.
- Absentmindedly posting "Guardians of G" into a political satire, the language model proposed "Guardians of the Gumby Bears". I laughed out loud and turned it into a whimsical political party.

REGENERATE

Step 5: Generate More Specific Anecdotes

[personal_anecdote]:

While crafting a superhero novel, I typed "Guardians of G" intending to type "Guardians of the Galaxy" but my language model amusingly suggested "Guardians of the Gumby Bears". It sparked an idea for a humorous subplot.

prompt:

Please rewrite 3 versions of this anecdote to make it more specific. Replace all generic terms with concrete and specific instances: [personal_anecdote].

GENERATE SPECIFIC ANECDOTES

Here are 3 specific versions:

- While I was diligently typing up my latest Batman-inspired book, my fingers swiftly moved to type the familiar phrase "Guardians of Gotham City." However, my cheeky autocorrected spellcheck unexpectedly suggested "Guardians of the Gumby Machine." This surprise aroused an intriguing concept for a side-splitting sideline to my main plot.
- Once, during an intense session of composing my Marvel-based masterpiece, I began to write "Guardians of the Cosmic Cosmos," calling back to my cherished "Guardians of the Galaxy" characters. But my computer, with its AI language model, hilariously replaced it with "Guardians of the Candy Cane!" This hilarious mix-up sparked a brilliant idea for an uproariously funny tangent within my main story.
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REGENERATE

Step 6: Generate Example Hooks

[specific_anecdote]:

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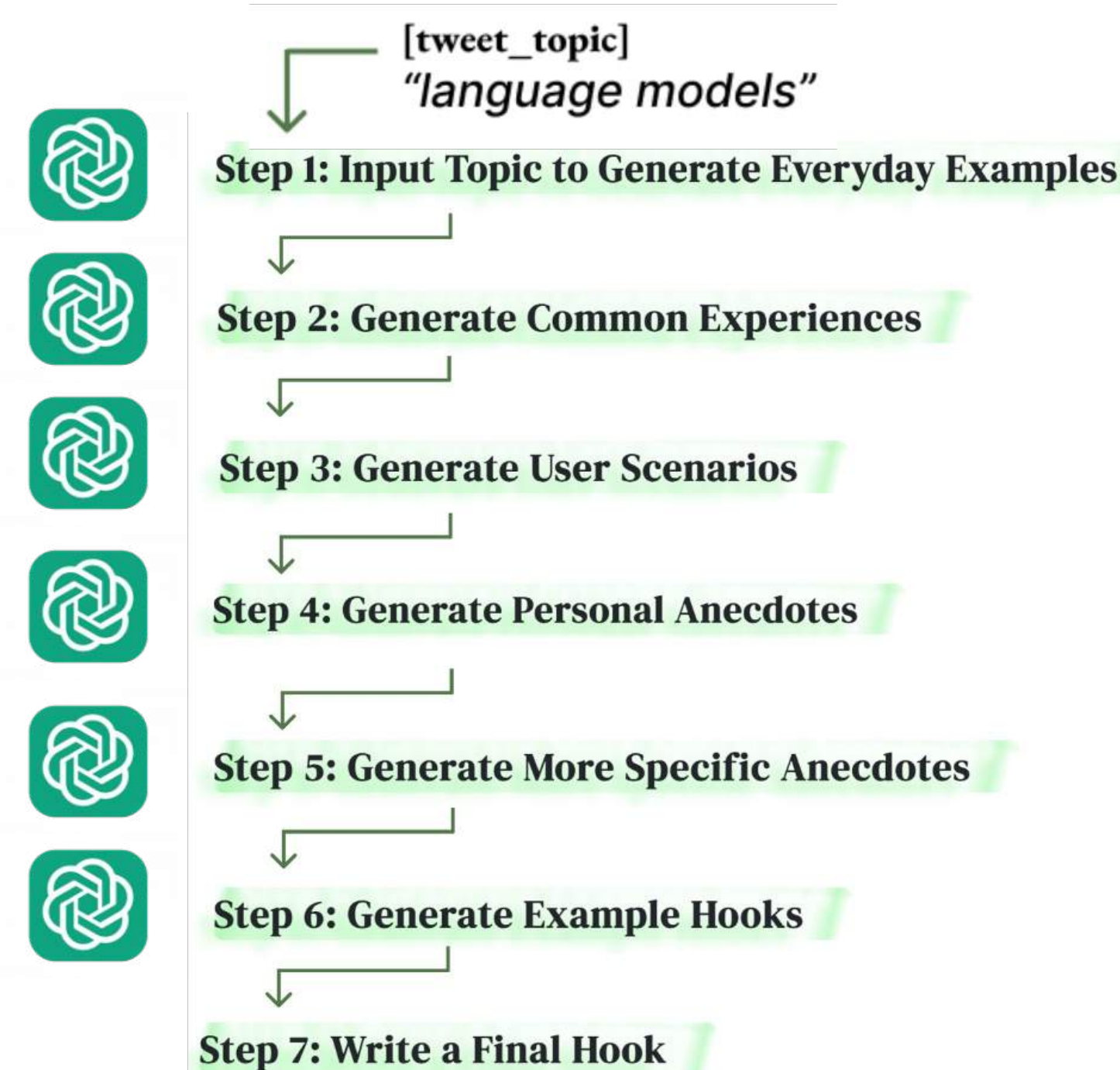
prompt:

Here is a personal anecdote about the topic, [tweet_topic], for the experience of [common_experience]. [specific_anecdote]. Based on this anecdote, please write 3 versions of the tweet hook to introduce the concept of [tweet_topic]. The tweet should have no jargon, it should contain a specific and relatable example, it should have a driving question to spark curiosity and make readers want to read more. Make sure to include a few emojis and all the details.

GENERATE EXAMPLE HOOKS!

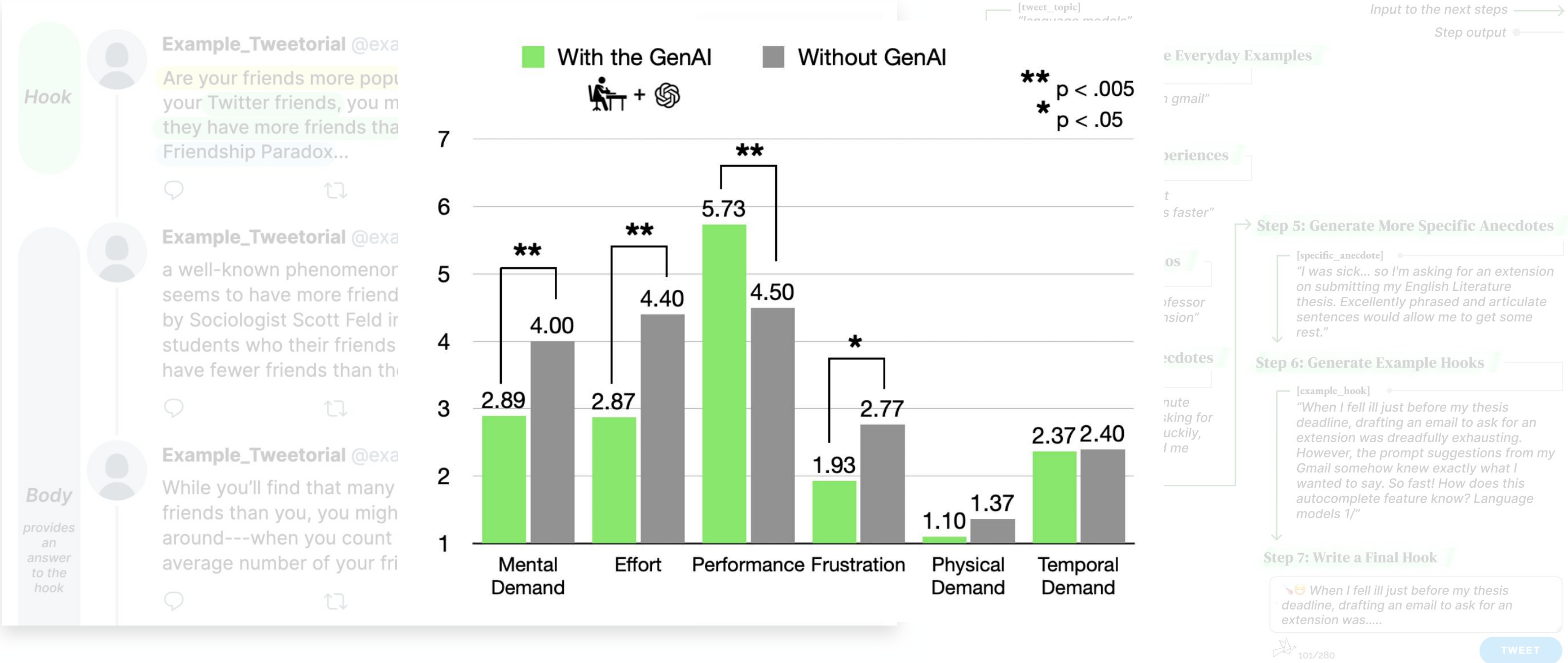
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- As I penned down an epic sci-fi novel last week, my writing assistant threw a curveball by suggesting "Guardians of the Gumby Bears" 😂 Can you believe it? I made me chuckle as well as ponder. So, what's generating these unexpected yet fun suggestions? Venture with me into the realm of "Language Models"! 📝
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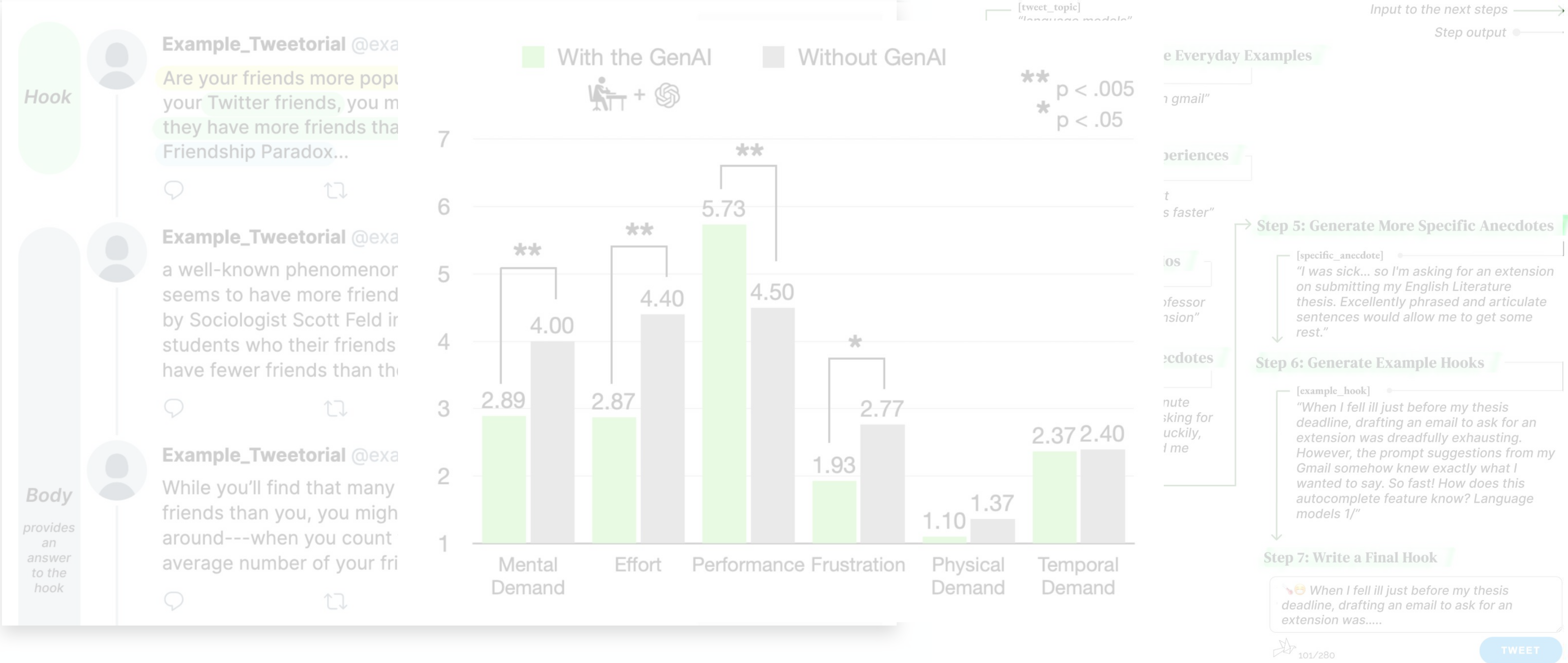


🤢🤒 When I fell ill just before my thesis deadline, drafting an email to ask for an

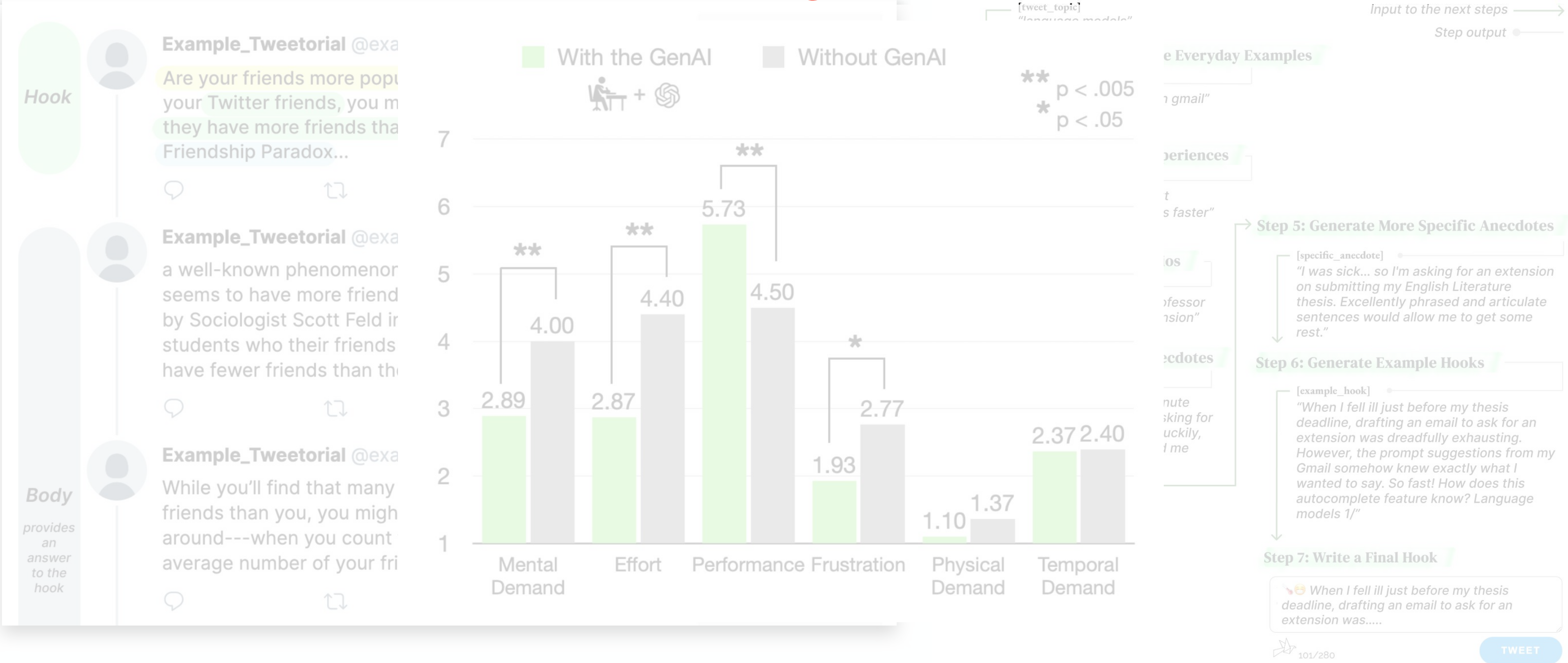
Generative AI workflows brings novel & impressive abilities as they chain AI outputs together with human interaction, {and they do it really well}



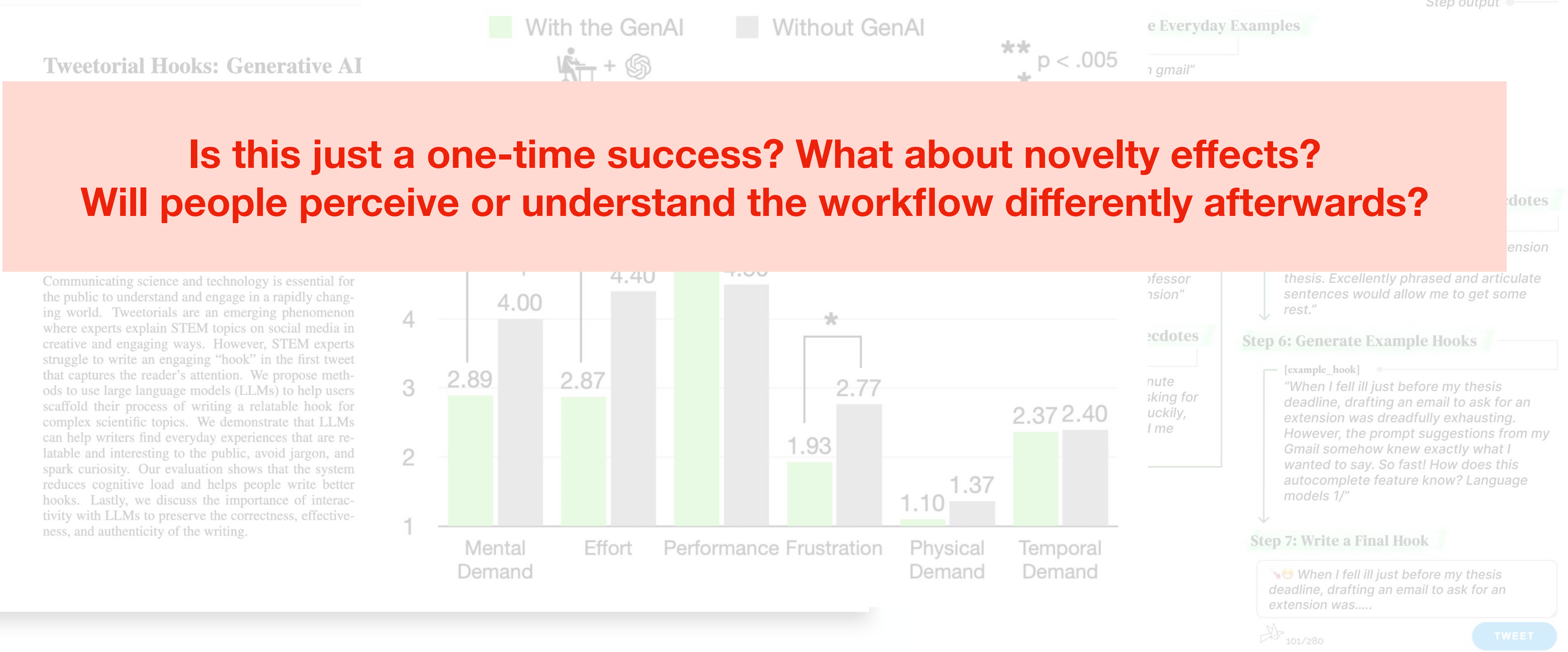
Generative AI workflows brings novel & impressive abilities as they chain AI outputs together with human interaction, {and they do it really well}



Generative AI workflows brings novel & impressive abilities as they chain AI outputs together with human interaction, {and they do it really well in long term??}



Generative AI workflows brings novel & impressive abilities as they chain AI outputs together with human interaction, {and they do it really well in long term??}



Generative AI workflows brings novel & impressive abilities as they chain AI outputs together with human interaction, {and they do it really well in long term??}

Is this just a one-time success? What about novelty effects?
Will people perceive or understand the workflow differently afterwards?



Some technologies are ***abandoned*** after the novelty wears off.

Activity Trackers (ex. fitbit)
(Sadeghi. et al., 2022)



But some get ***more useful*** over time. Familiarization actually makes it more useful.

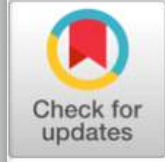
Gamification in Education
(Rodrigues. et al., 2022)



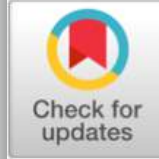
Step 7: Write a Final Hook

When I fell ill just before my thesis deadline, drafting an email to ask for an extension was.....

TWEET



Not Just Novelty: A Longitudinal Study on Utility and Customization of an AI Workflow

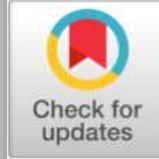


Not Just Novelty: A Longitudinal Study on Utility and Customization of an AI Workflow

RQ 1: Familiarization

How long is the familiarization phase? What happens around it?

How do the usefulness and performance of the workflow change before and after familiarization?



Not Just Novelty: A Longitudinal Study on Utility and Customization of an AI Workflow

RQ 1: Familiarization

**How long is the familiarization phase? What happens around it?
How do the usefulness and performance of the workflow change before and after familiarization?**

RQ 2: Customization

How and why do users customize the workflow?



Not Just Novelty: A Longitudinal Study on Utility and Customization of an AI Workflow

RQ 1: Familiarization

**How long is the familiarization phase? What happens around it?
How do the usefulness and performance of the workflow change before and after familiarization?**

RQ 2: Customization

How and why do users customize the workflow?

RQ 3: Mental Models and Ownership

How have users' mental models, involment, and ownership changed over time?

Method:

- **3-week deployment study on the co-creation of Tweetorial hooks**
- **12 STEM experts**
- **10 sessions of system usage**

Step 1: Input Topic to Generate Everyday Examples



Step 2: Generate Common Experiences



Step 3: Generate User Scenarios



Step 4: Generate Personal Anecdotes



Step 5: Generate More Specific Anecdotes



Step 6: Generate Example Hooks



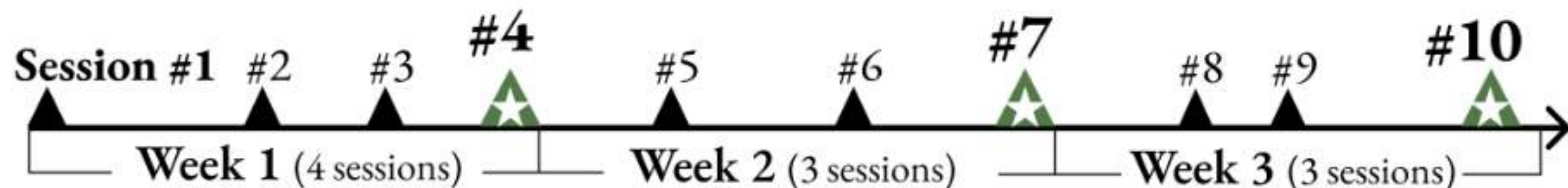
Step 7: Write a Final Hook

Method:

- 3-week deployment study on the co-creation of Tweetorial hooks
- 12 STEM experts
- 10 sessions of system usage,
 - survey after each usage
 - Interview every week

▲ System Usage + Survey

▲ System Usage + Survey + Interview



GenAI workflows are easy to customize. Will users do it?

[common_experience]:

Unexpectedly humorous or nonsensical suggestions.

GENERATE USER SCENARIOS

Here are 5 user scenarios:

1. While writing an email, it suggested I end with "Roar like a dinosaur."

2. It replaced "Good morning" with "Party with unicorns."

3. It suggested to rephrase "Will you marry me?" with "Can you be my penguin?"

4. Using it as a recipe guide, it added "dance salsa for 5 mins" in cooking steps.

5. It replaced "meeting at 3pm" with "mammoth cave tour at sunset."

6. When typing "Guardians of", it completed as "Guardians of the gummy bears."

7. While typing a complaint letter, it suggested "Let's have a pizza party."

8. Asked about the weather, it replied me with "Cloudy chance of flying saucers."

9. Documenting meeting minutes, it recommended "Meeting was an alien conspiracy."

Hidden prompts

Original Interface

GenAI workflows are easy to customize. Will users do it?

[common_experience]:

Unexpectedly humorous or nonsensical suggestions.

prompt:

Please list 12 user scenarios someone encounters with [common_experience] while using [tweet_topic]. Each scenario should be under 15 words, from a first-person perspective, related to a general audience, and not require technical knowledge.

GENERATE USER SCENARIOS

Here are 5 user scenarios:

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7. While typing a complaint letter, it suggested "Let's have a pizza party."

8. Asked about the weather, it replied me with "Cloudy chance of flying saucers."

9. Documenting meeting minutes, it recommended "Meeting was an alien conspiracy."

prompt: NEW

Please list 12 user scenarios someone encounters with [common_experience] while using [tweet_topic]. Each scenario should be under 15 words, from a first-person perspective, related to a general audience, and not require technical knowledge.

Customizable Interface

GenAI workflows are easy to customize. Will users do it?

[common_experience]:

Unexpectedly humorous or nonsensical suggestions.

prompt:

Please list 12 user scenarios someone encounters with [common_experience] while using [tweet_topic]. Each scenario should be under 15 words, from a first-person perspective, related to a general audience, and not require technical knowledge.

GENERATE USER SCENARIOS

Here are 5 user scenarios:

- 1. While writing an email, it suggested I end with "Roar like a dinosaur."
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- 3. It suggested to rephrase "Will you marry me?" with "Can you be my penguin?"
- 4. Using it as a recipe guide, it added "dance salsa for 5 mins" in cooking steps.
- 5. It replaced "meeting at 3pm" with "mammoth cave tour at sunset."
- 6. When typing "Guardians of", it completed as "Guardians of the gummy bears." ↓ Copied to Next Step
- 7. While typing a complaint letter, it suggested "Let's have a pizza party."
- 8. Asked about the weather, it replied me with "Cloudy chance of flying saucers."
- 9. Documenting meeting minutes, it recommended "Meeting was an alien conspiracy."

prompt: NEW

Please list 12 user scenarios someone encounters with [common_experience] while using [tweet_topic]. Each scenario should be under 15 words, ~~from a first-person perspective,~~ related to a general audience, and not require technical knowledge.

GenAI workflows are easy to customize. Will users do it?

[common_experience]:

Unexpectedly humorous or nonsensical suggestions.

prompt:

Please list 12 user scenarios someone encounters with [common_experience] while using [tweet_topic]. Each scenario should be under 15 words, from a first-person perspective, related to a general audience, and not require technical knowledge.

GENERATE USER SCENARIOS

Here are 5 user scenarios:

- 1. While writing an email, it suggested I end with "Roar like a dinosaur."
- 2. It replaced "Good morning" with "Party with unicorns."
- 3. It suggested to rephrase "Will you marry me?" with "Can you be my penguin?"
- 4. Using it as a recipe guide, it added "dance salsa for 5 mins" in cooking steps.
- 5. It replaced "meeting at 3pm" with "mammoth cave tour at sunset."
- 6. When typing "Guardians of", it completed as "Guardians of the gummy bears." ↓ Copied to Next Step
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prompt: NEW

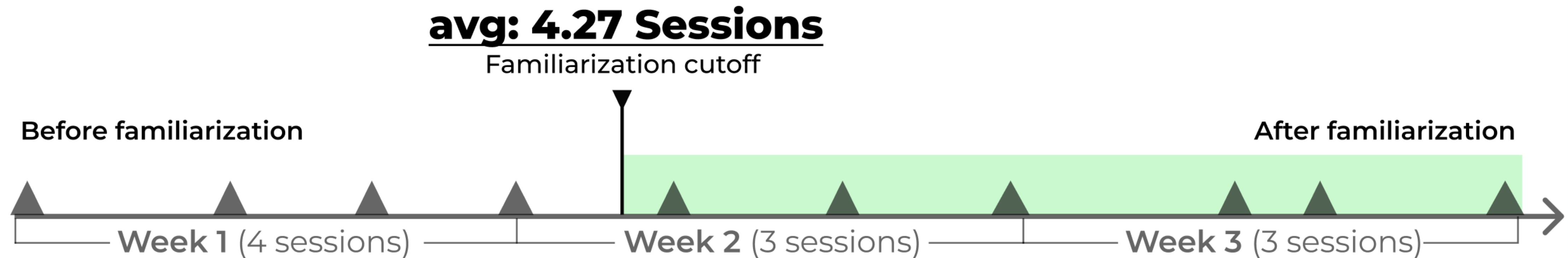
Please list 12 user scenarios someone encounters with [common_experience] while using [tweet_topic]. Each scenario should be under 15 words, ~~from a first-person perspective,~~ related to a general audience, and not require technical knowledge. Add emoji

Finding

Finding 1: Familiarization Time

Takeaway: People need at least a few tries to familiarize.

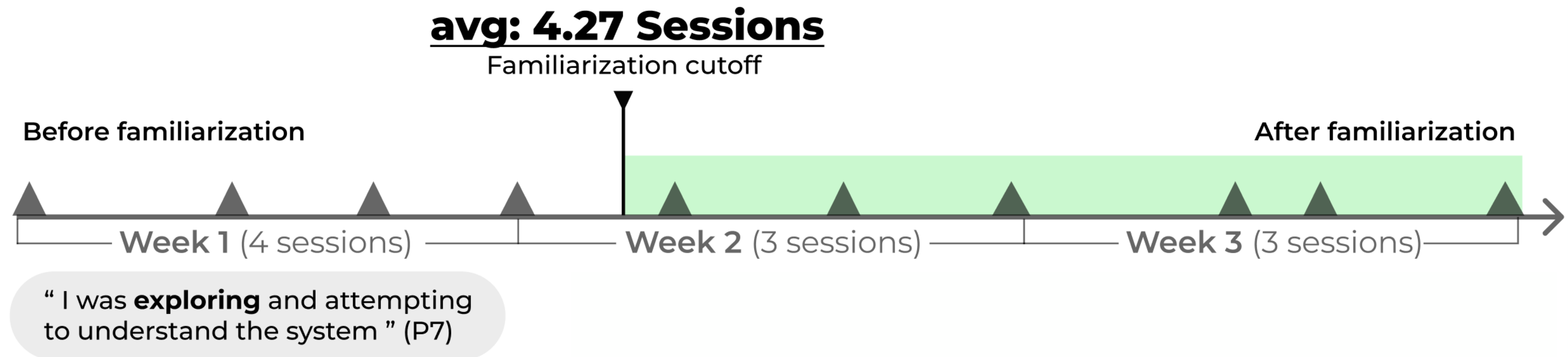
Familiarization phase lasts avg. ***4.27*** sessions $\approx$ 1.3 weeks $\approx$ 1.42 hour.



Finding 1: Familiarization Time

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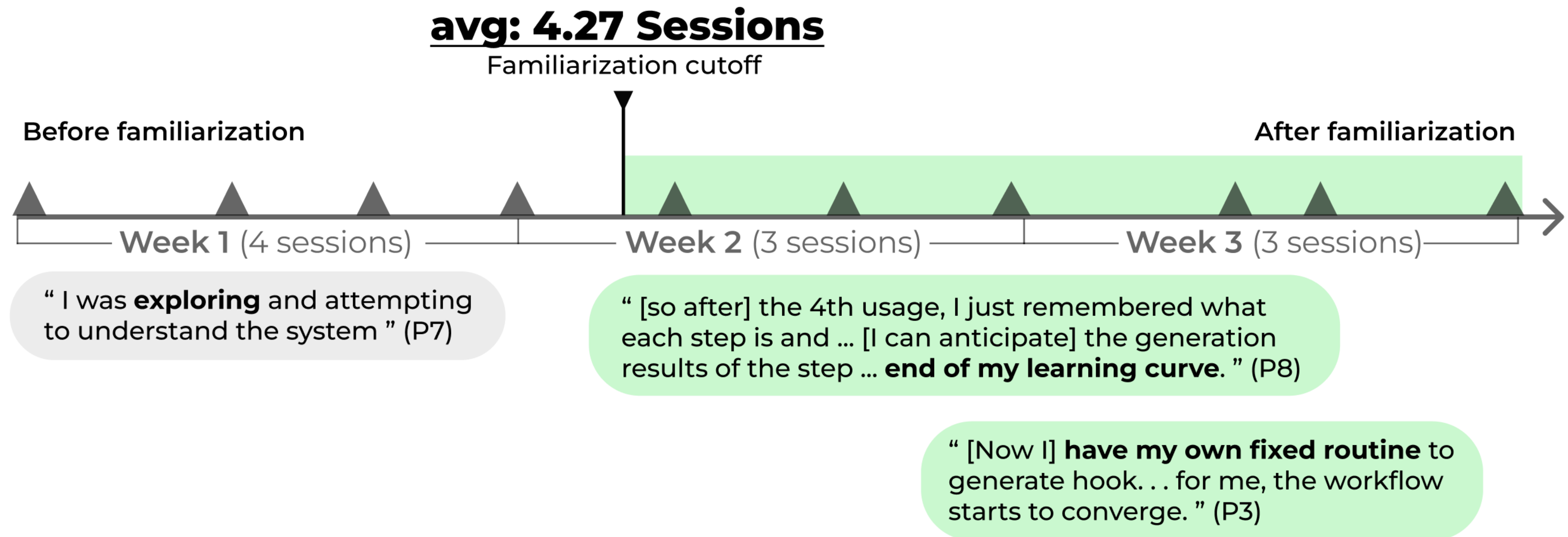
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Finding 1: Familiarization Time

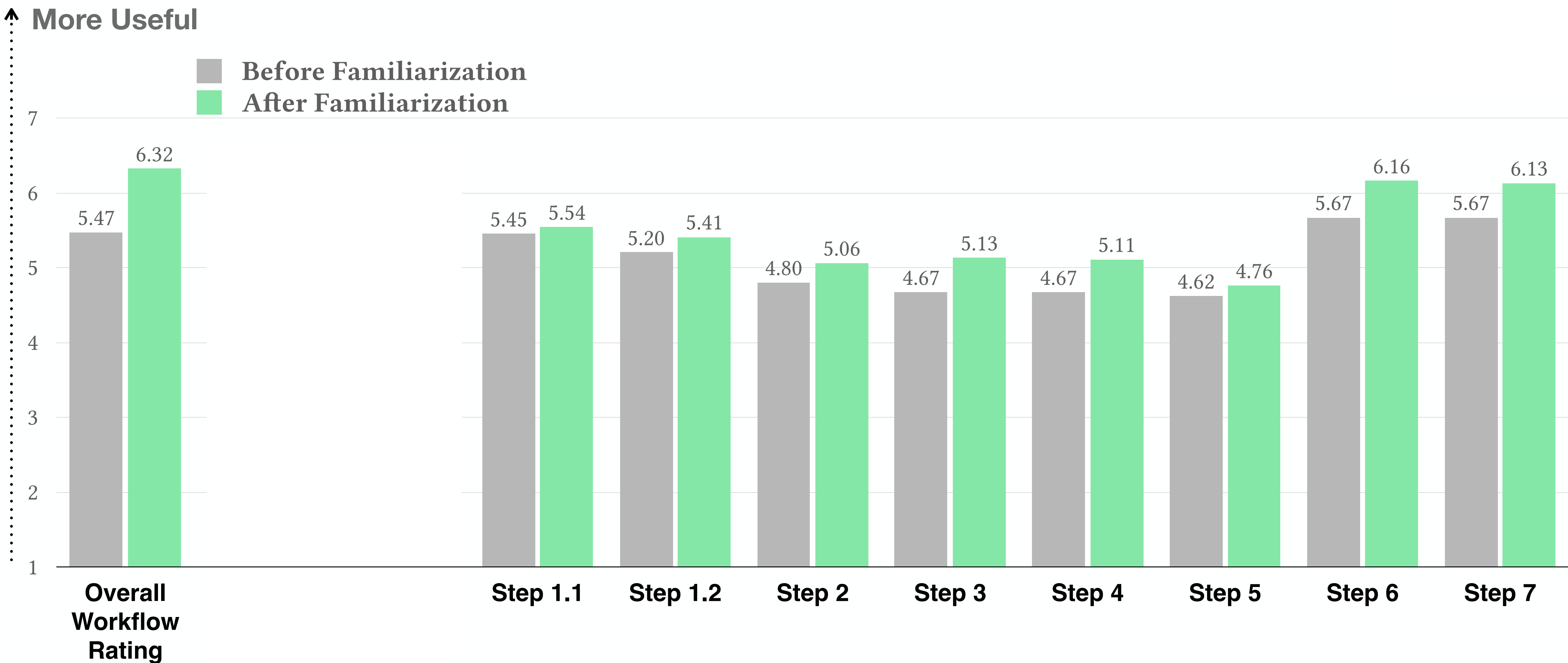
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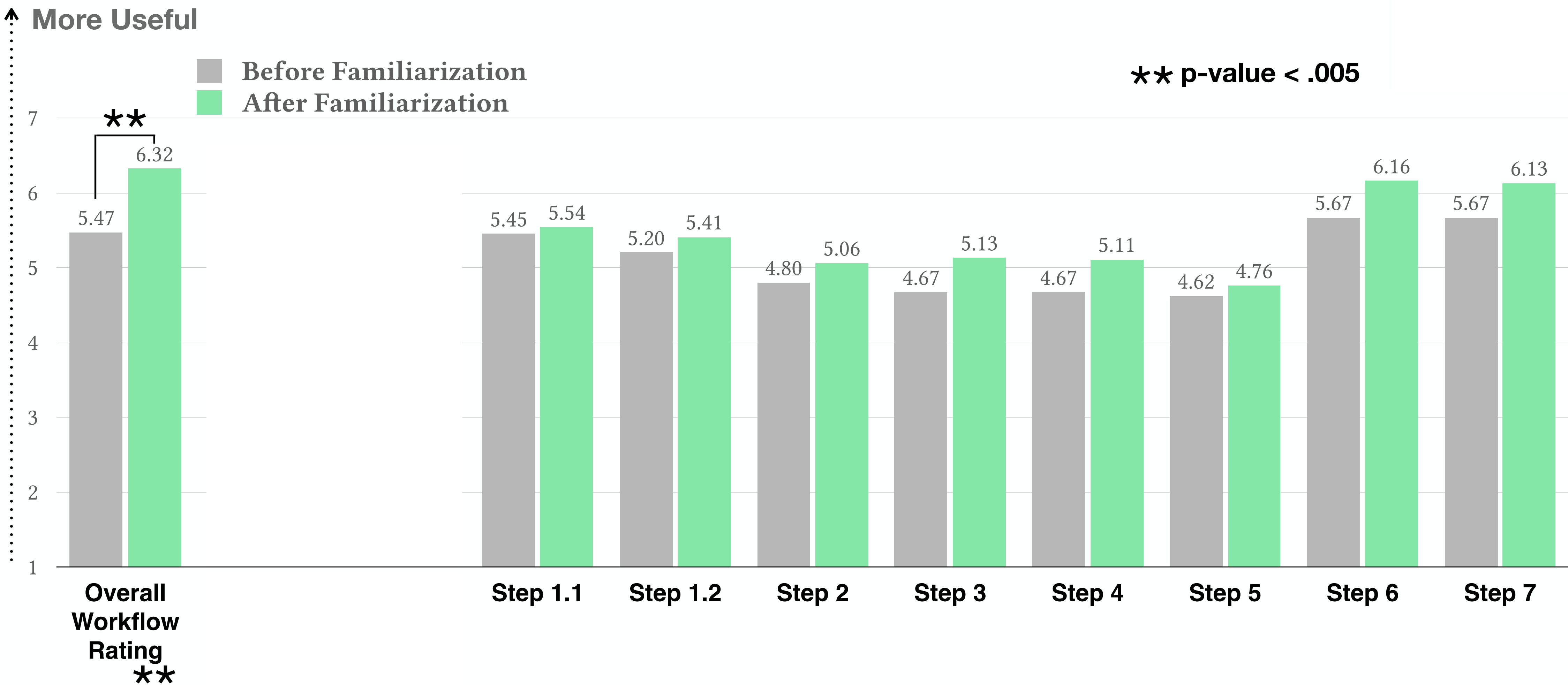
Finding 2: Changes in Workflow Utility

Takeaway: Overall workflow rated more useful after familiarization



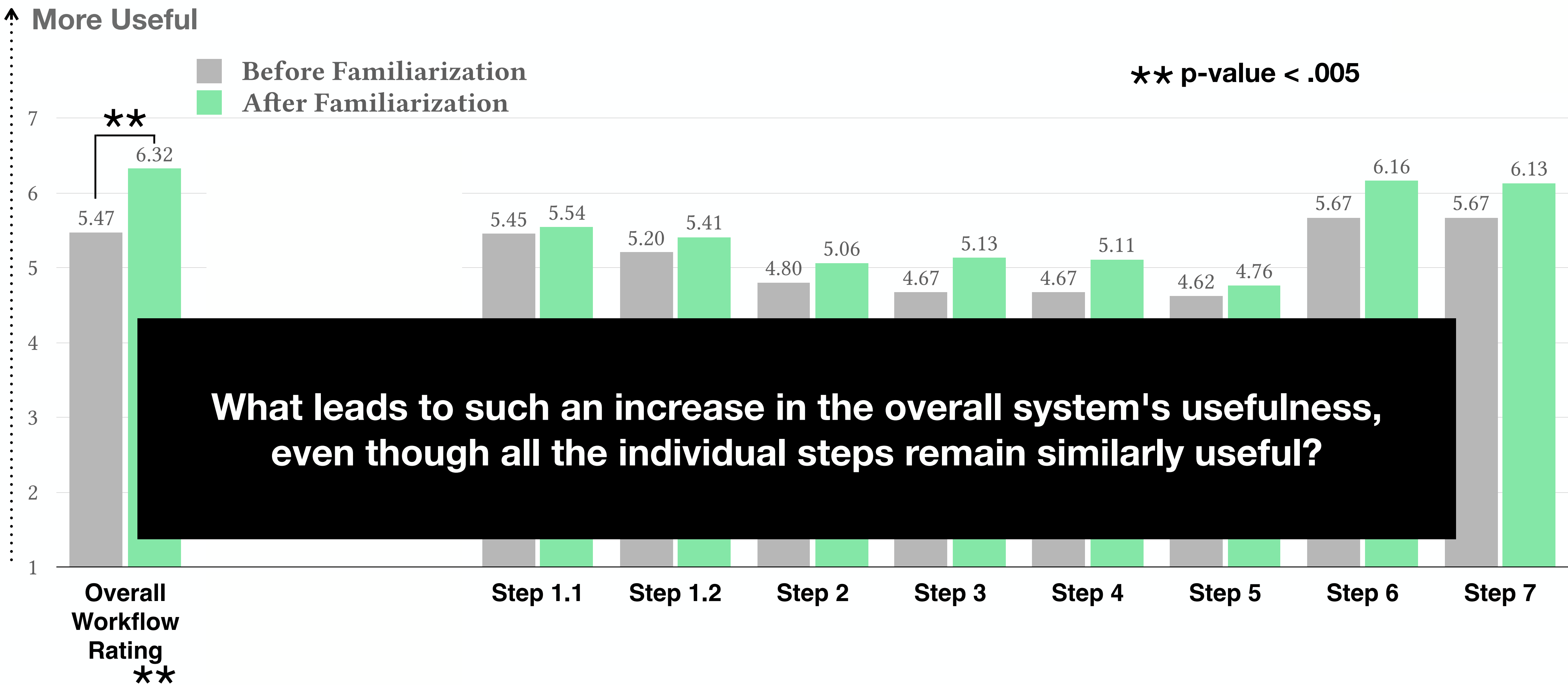
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Takeaway: Overall workflow rated more useful after familiarization



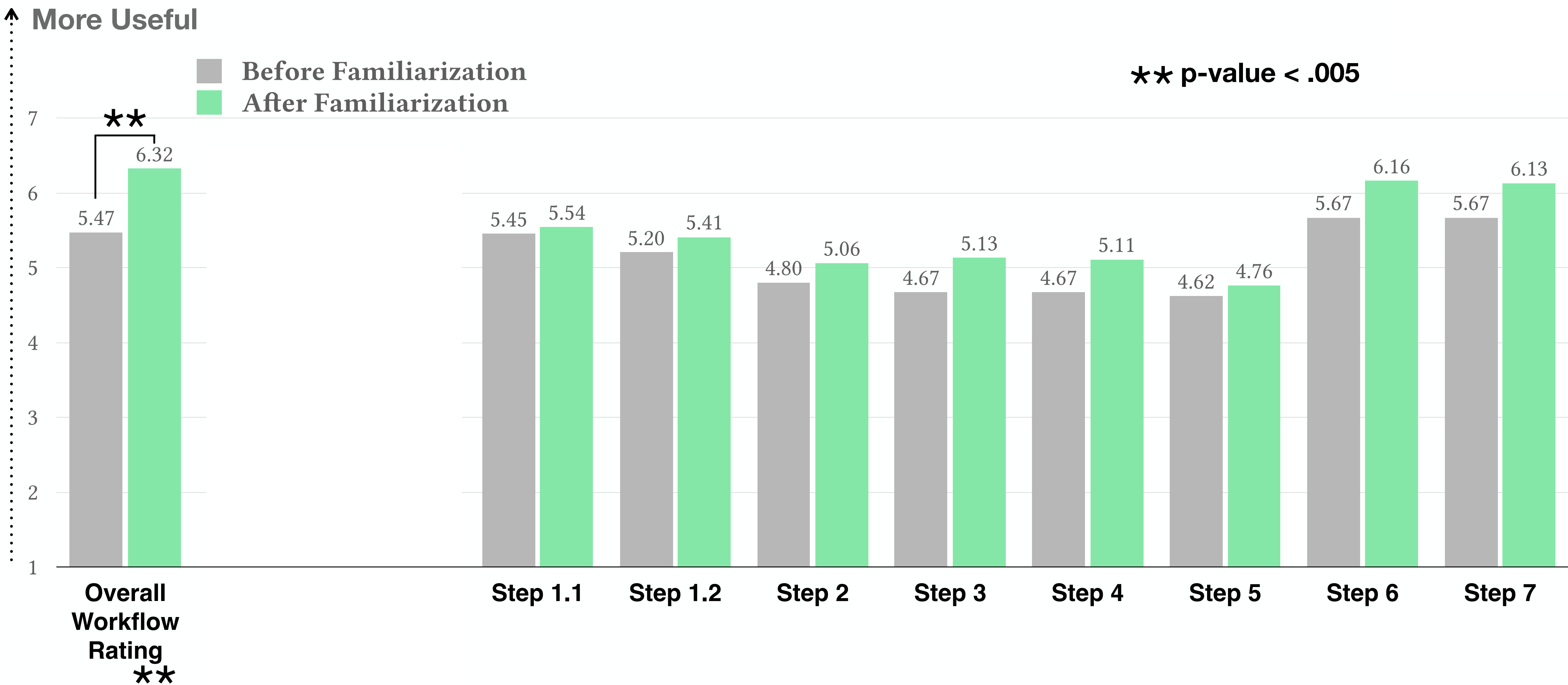
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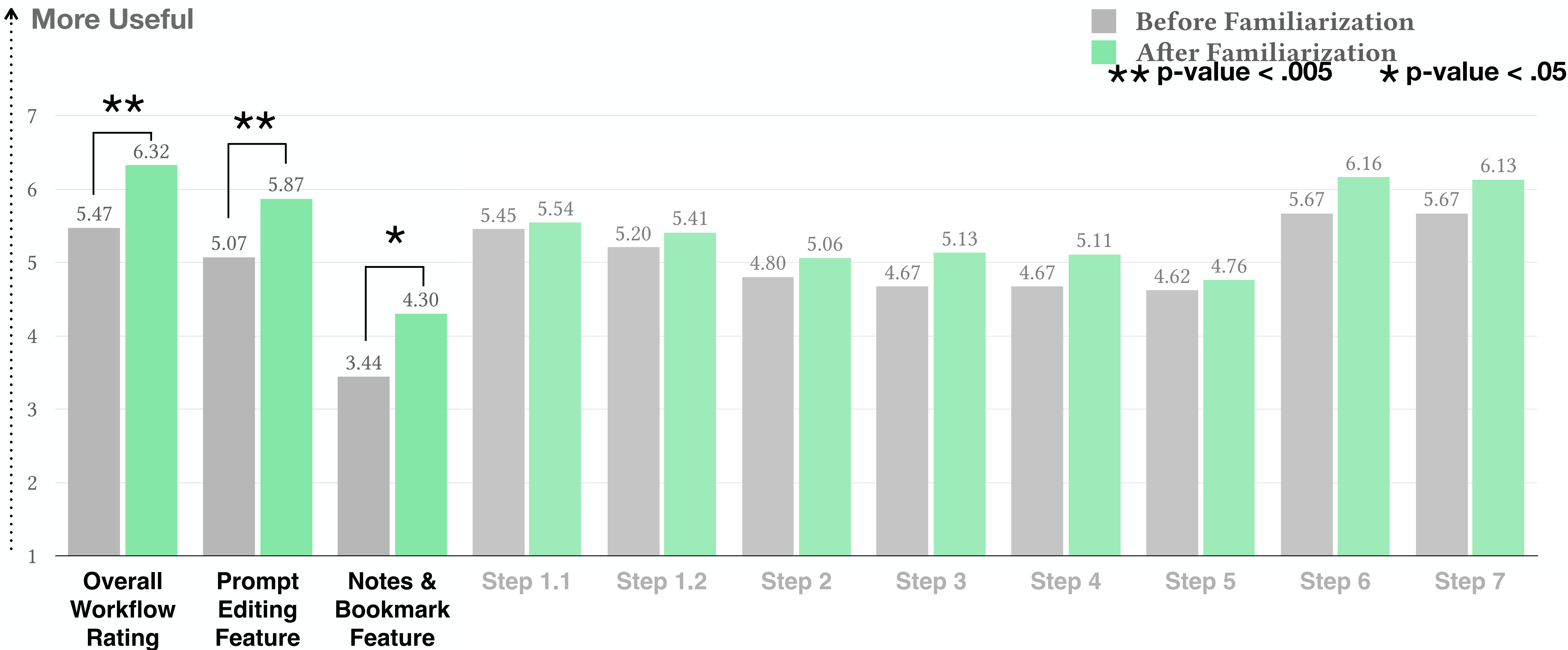
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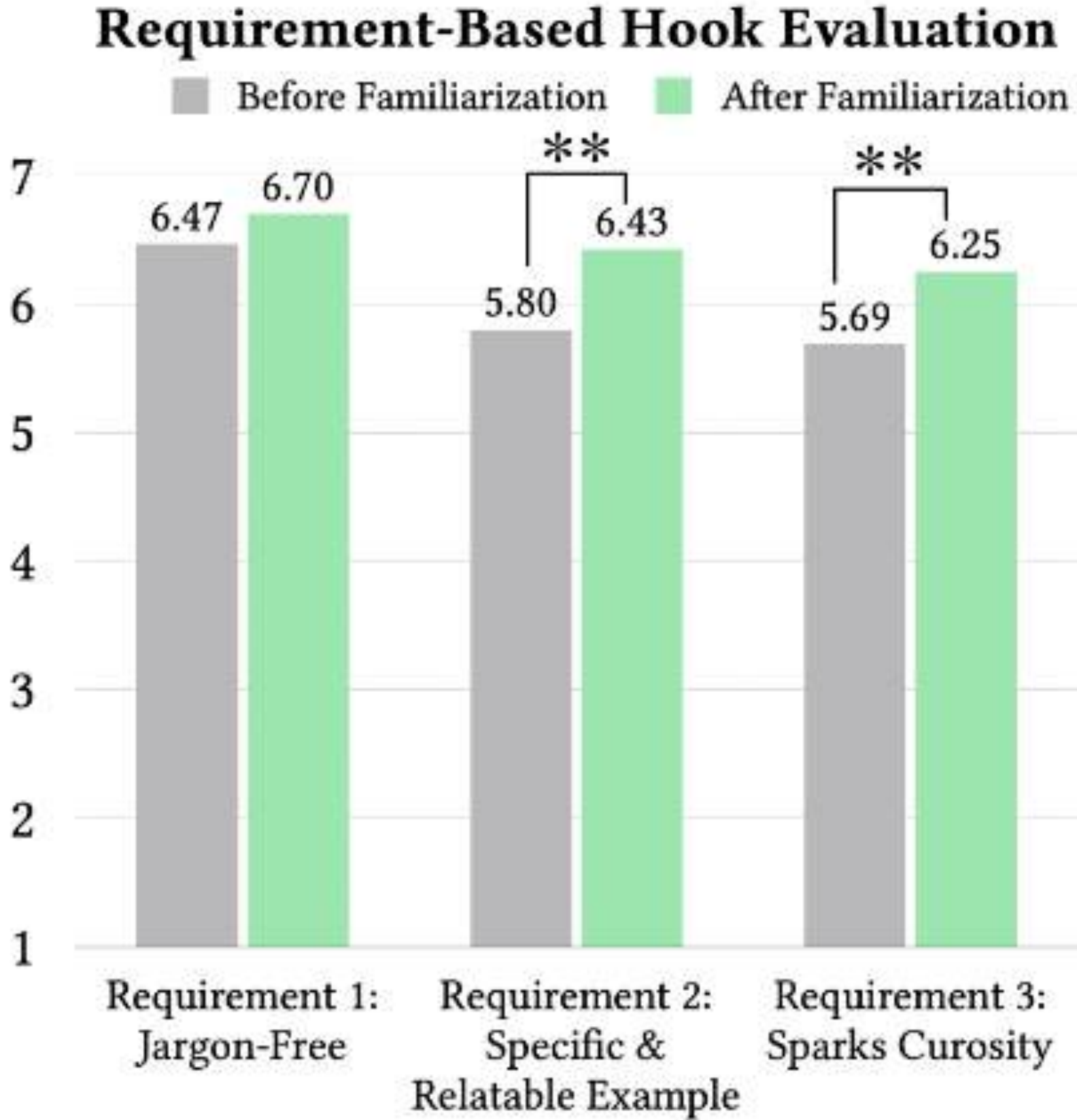
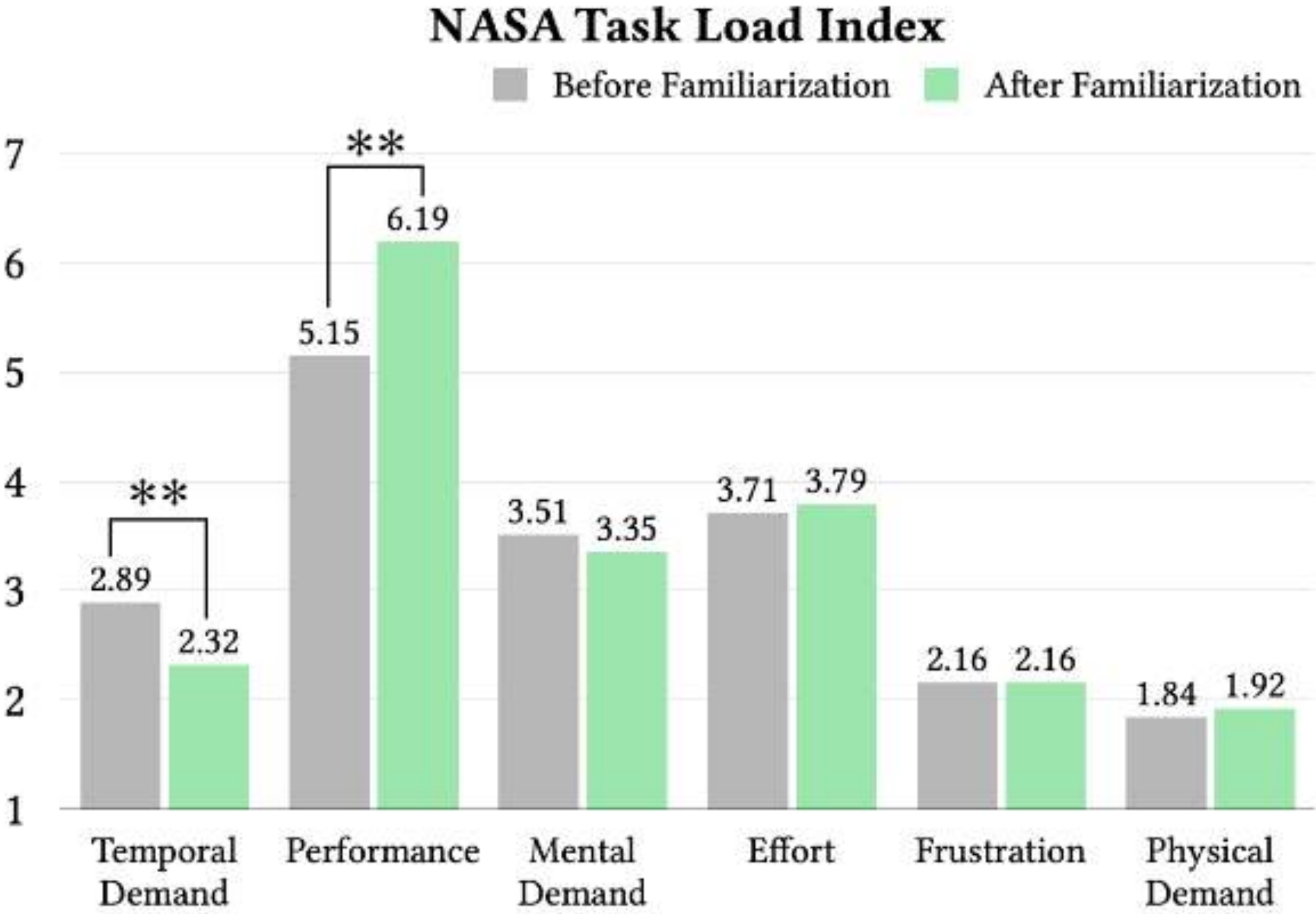


Finding 2: Changes in Workflow Utility

Takeaway: Overall workflow rated more useful after familiarization
AND, the success of customization leads to increased usefulness.



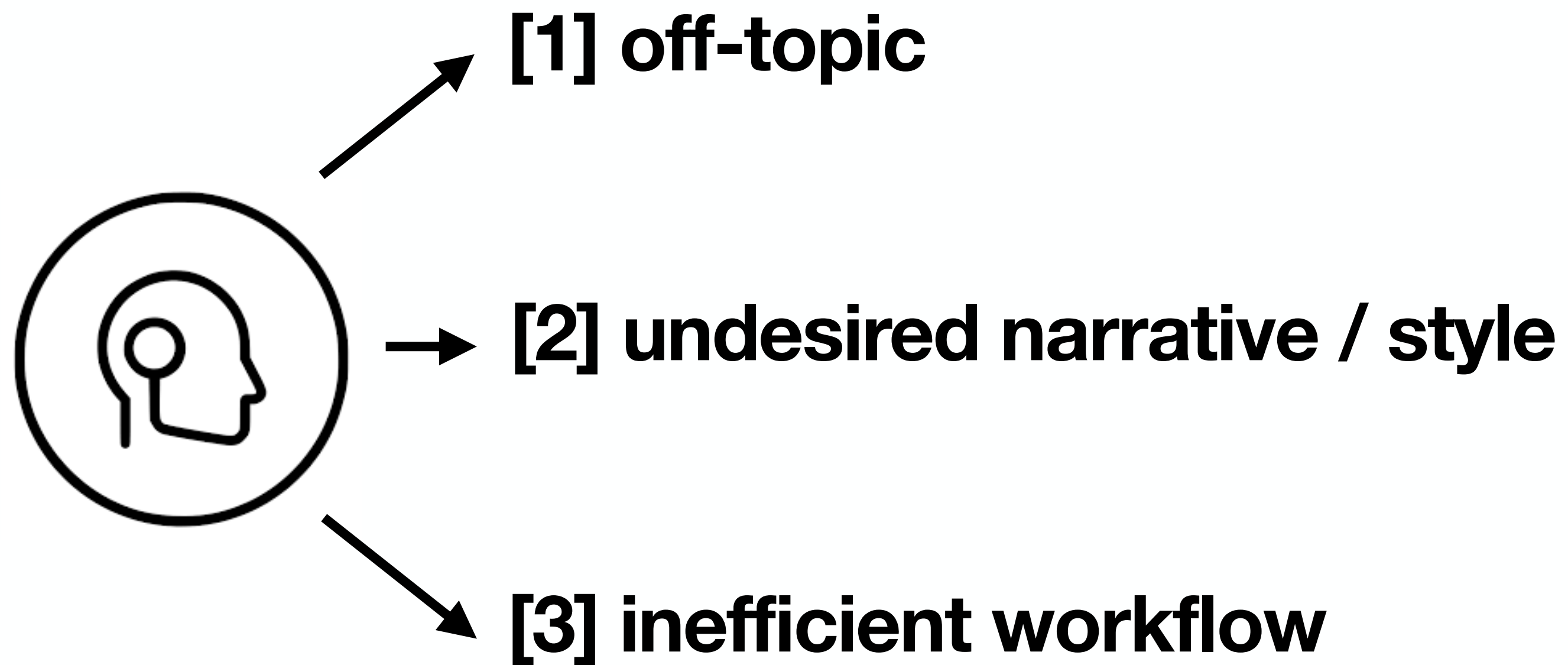
Finding 2: Workflow rated more useful after familiarization & after people know how to customize. The success of customization contributes to the increased usefulness. & better user performance over the task!



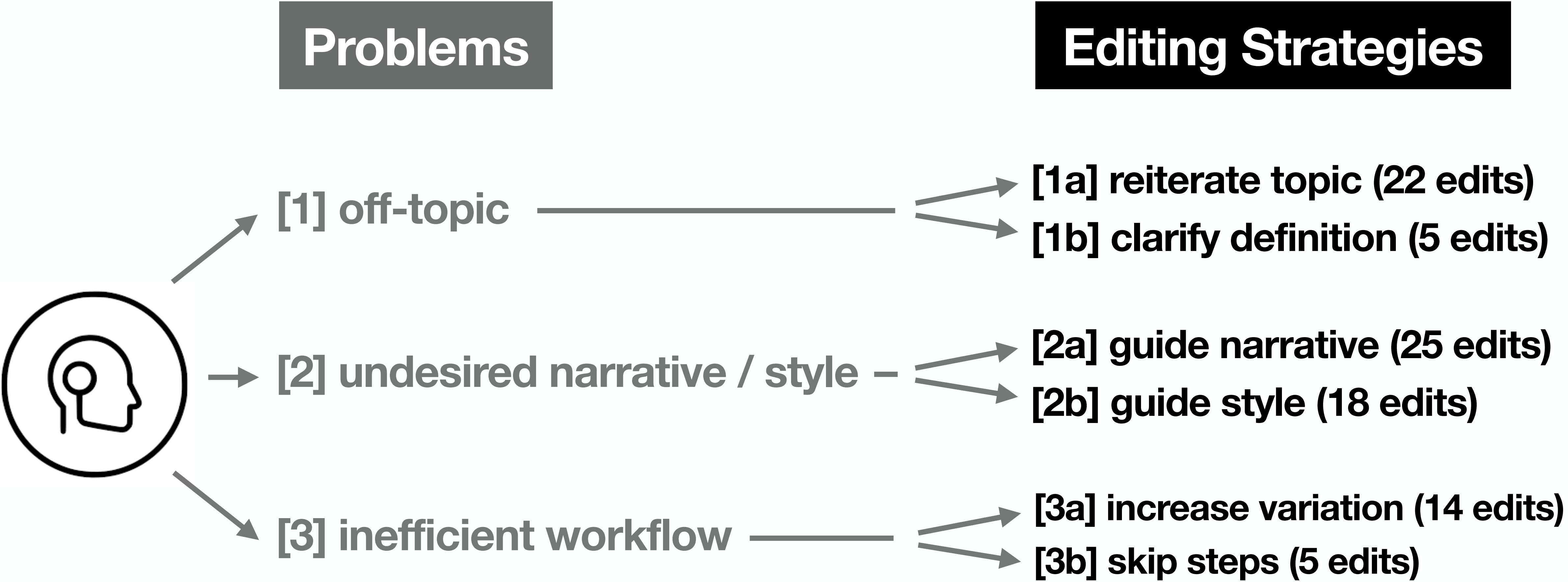
Finding 3: How Users Customize.

Finding 3: How Users Customize.

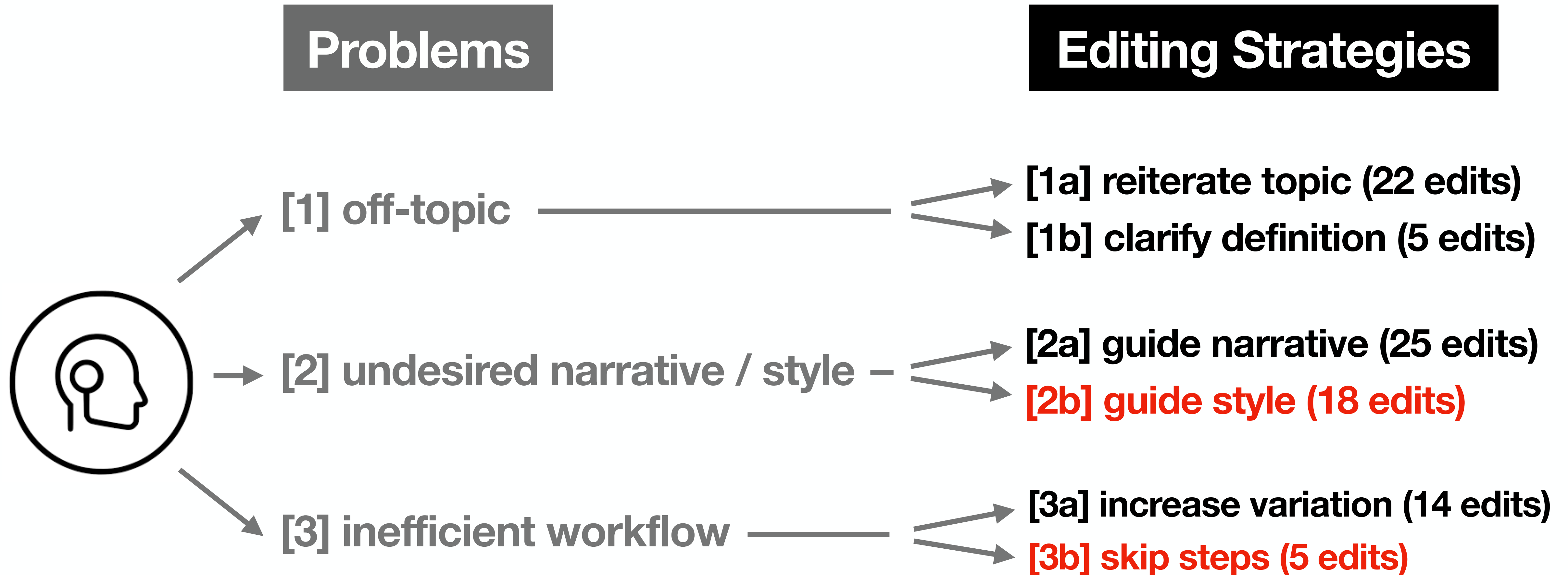
Problems



Finding 3: How Users Customize.



Finding 3: How Users Customize.



Finding 4: Involvement & Ownership Over Time

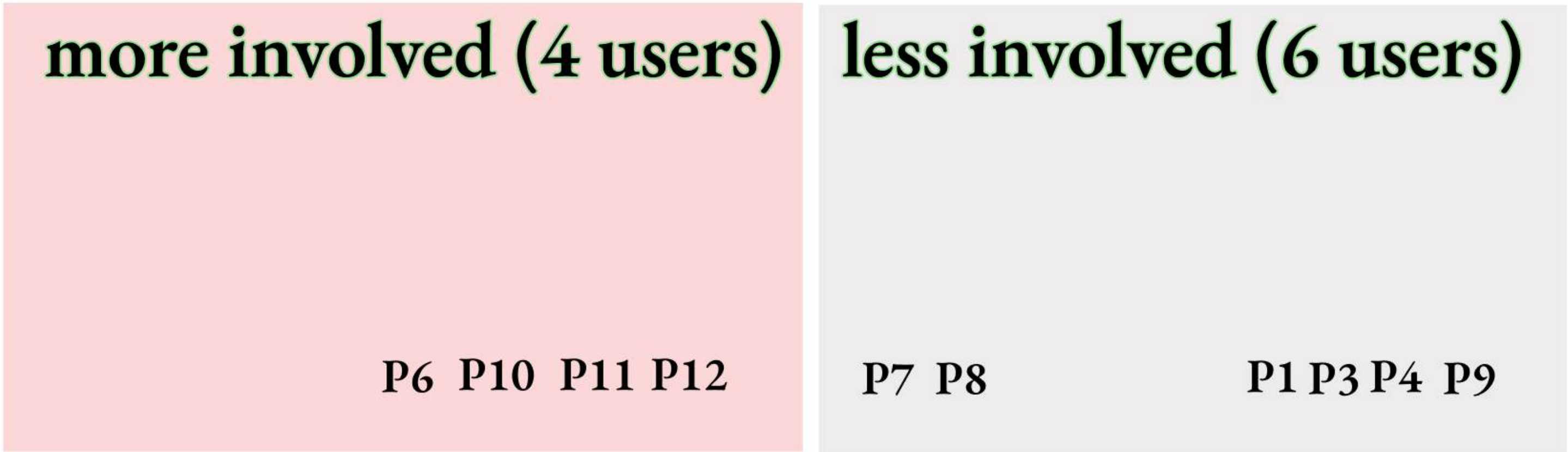
Overtime, I felt I became...

more involved (4 users)

P6 P10 P11 P12

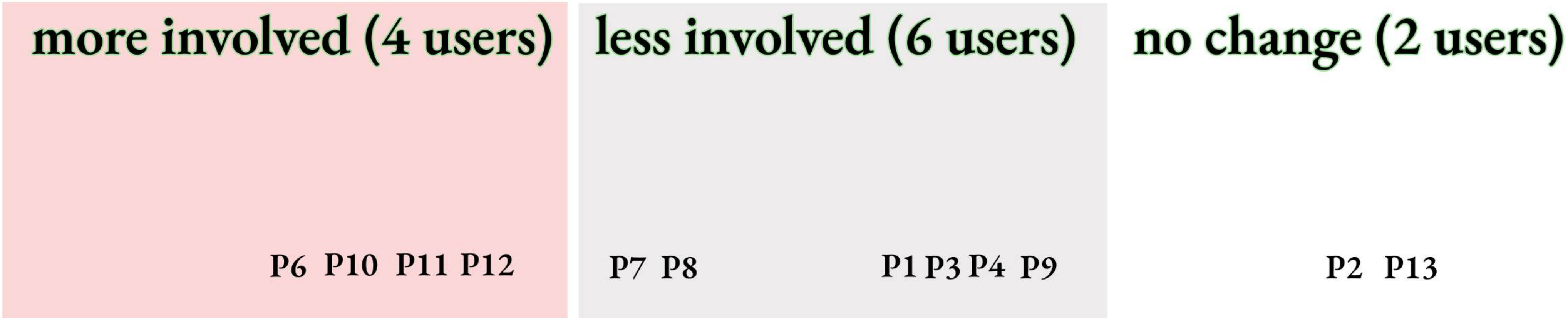
Finding 4: Involvement & Ownership Over Time

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Finding 4: Involvement & Ownership Over Time

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Finding 4: Involvement & Ownership

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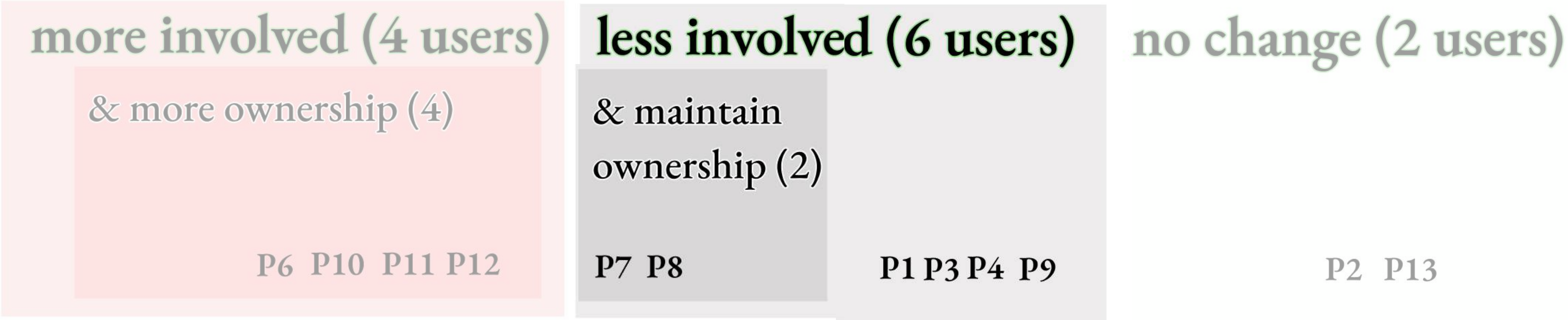
More involved, more ownership: "My hard work pays off"

"After making more edits and being more engaged in the interaction, I feel a stronger connection to the tone and effort."

— P10, Week 3 Interview

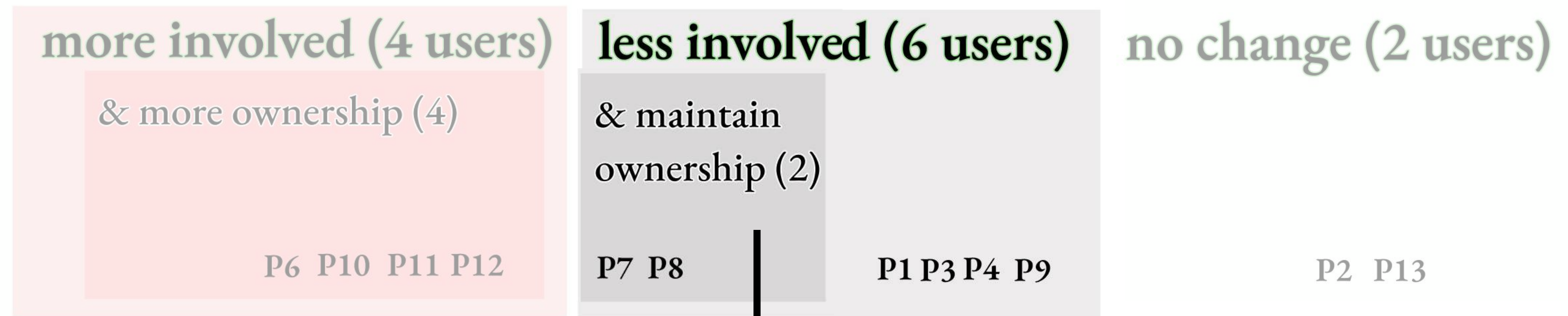
Finding 4: Involvement & Ownership

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Finding 4: Involvement & Ownership

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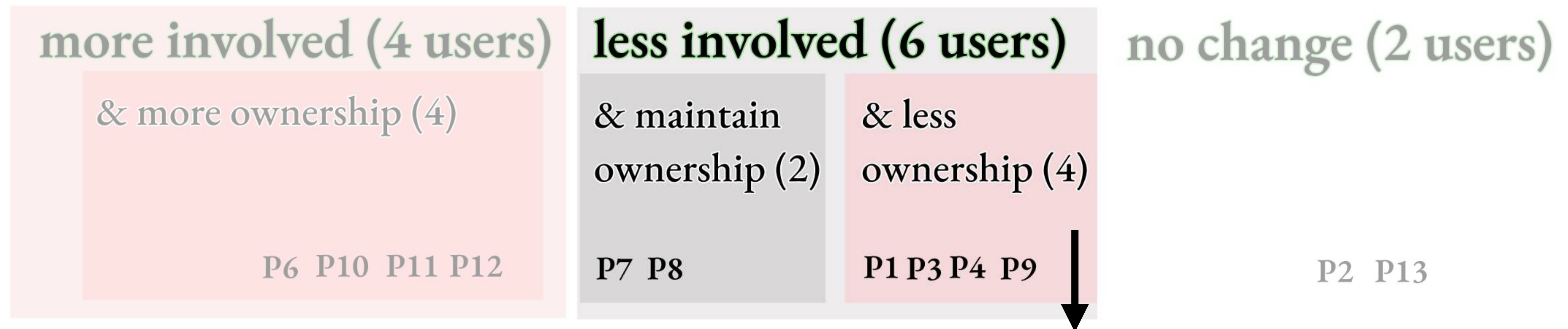
Less involved, but maintain ownership. "I still lead the system" (2 users).

"I feel like, even if I did no revision at all, I'd still feel a high ownership because I did all the selections [and] I did guide the system through... but still it's overall pretty high ownership generally."

— P8, Week 3 Interview

Finding 4: Involvement & Ownership

Overtime, I felt I became...



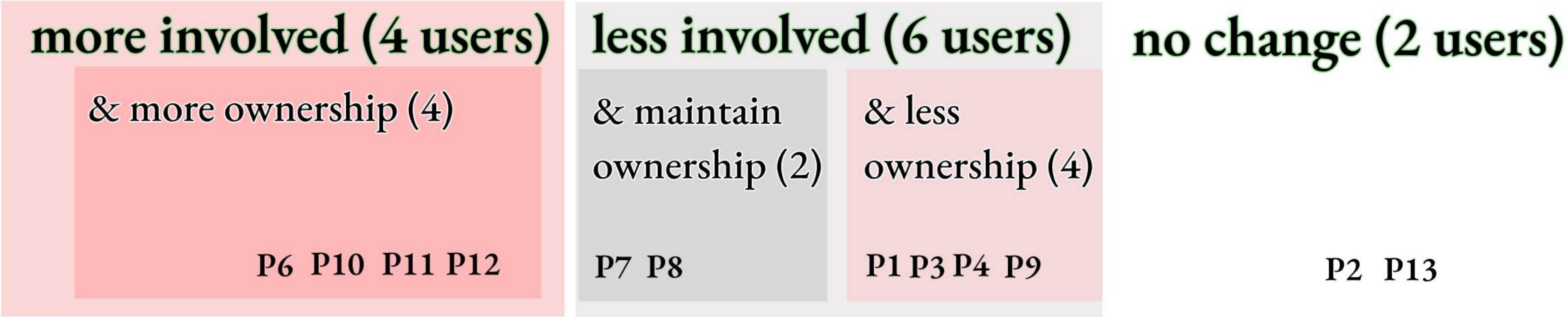
Less involved, less ownership. "I am now just a reviewer"

"I know the system's capability better... I can trust its ability [and] become more reliant on the system... I become more like a human reviewer to help guide the GPT to generate a better hook... [so now] I think my ownership dropped a lot..."

— P3, Week 3 Interview

Finding 4: Involvement & Ownership

Overtime, I felt I became...





Not Just Novelty: A Longitudinal Study on Utility and Customization of an AI Workflow



Paper pdf

1 / Problem

Gen AI workflows bring novel abilities & **easy to customize**
But... is this due to the novelty effect?

RQ: What is the **long-term usefulness** of these AI tools?

2 / Three-Week Study (12 Users)

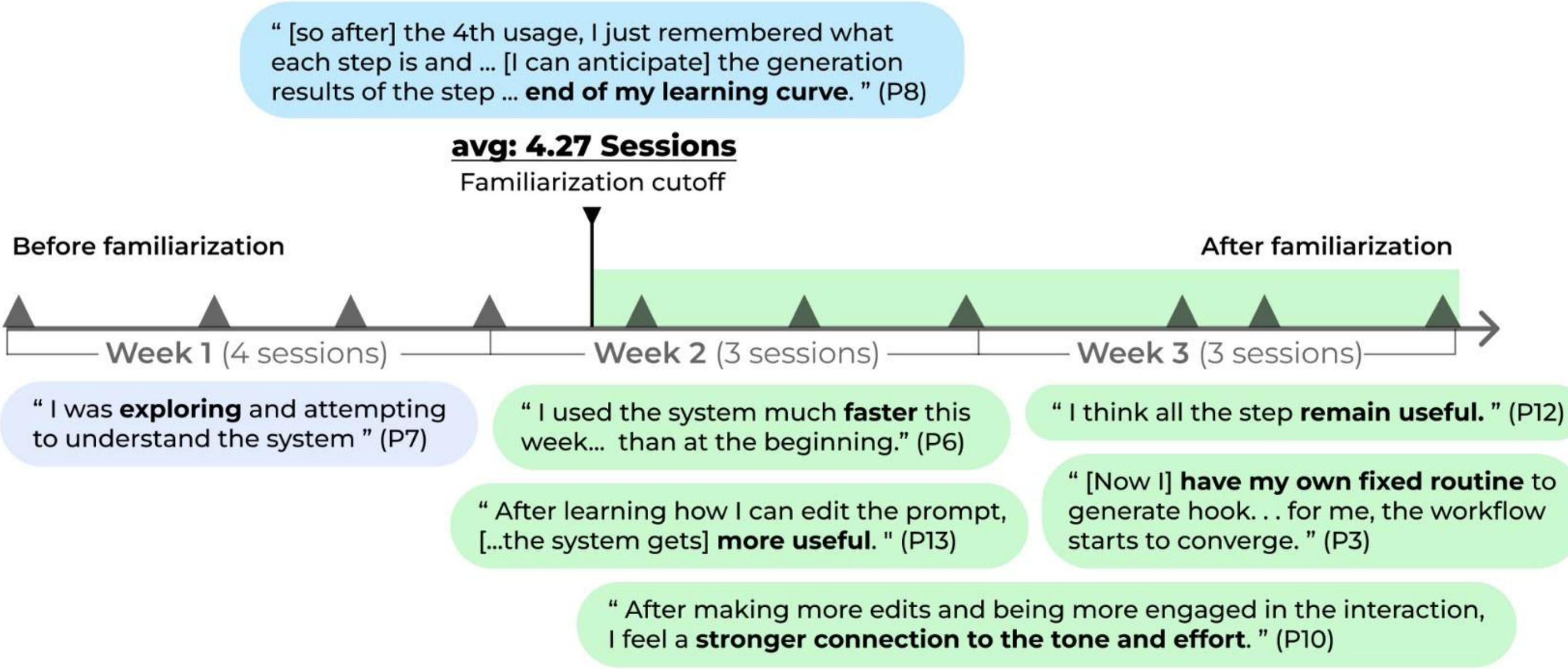
a proven useful AI workflow + customization (prompt edits)



3/ Findings

Not just novelty! Users find the system **more useful**** after familiarization, **as they can customize**** it in various ways, adapting to their needs.

** p-value < 0.005



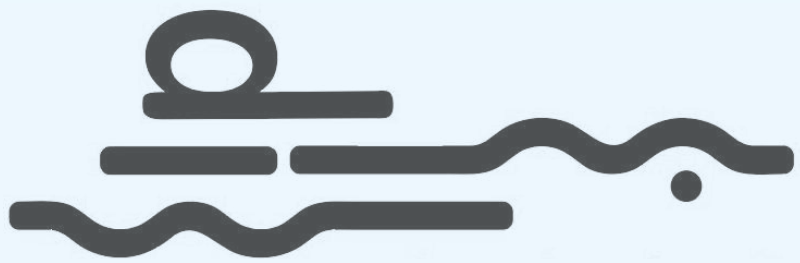
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Columbia HCI Group

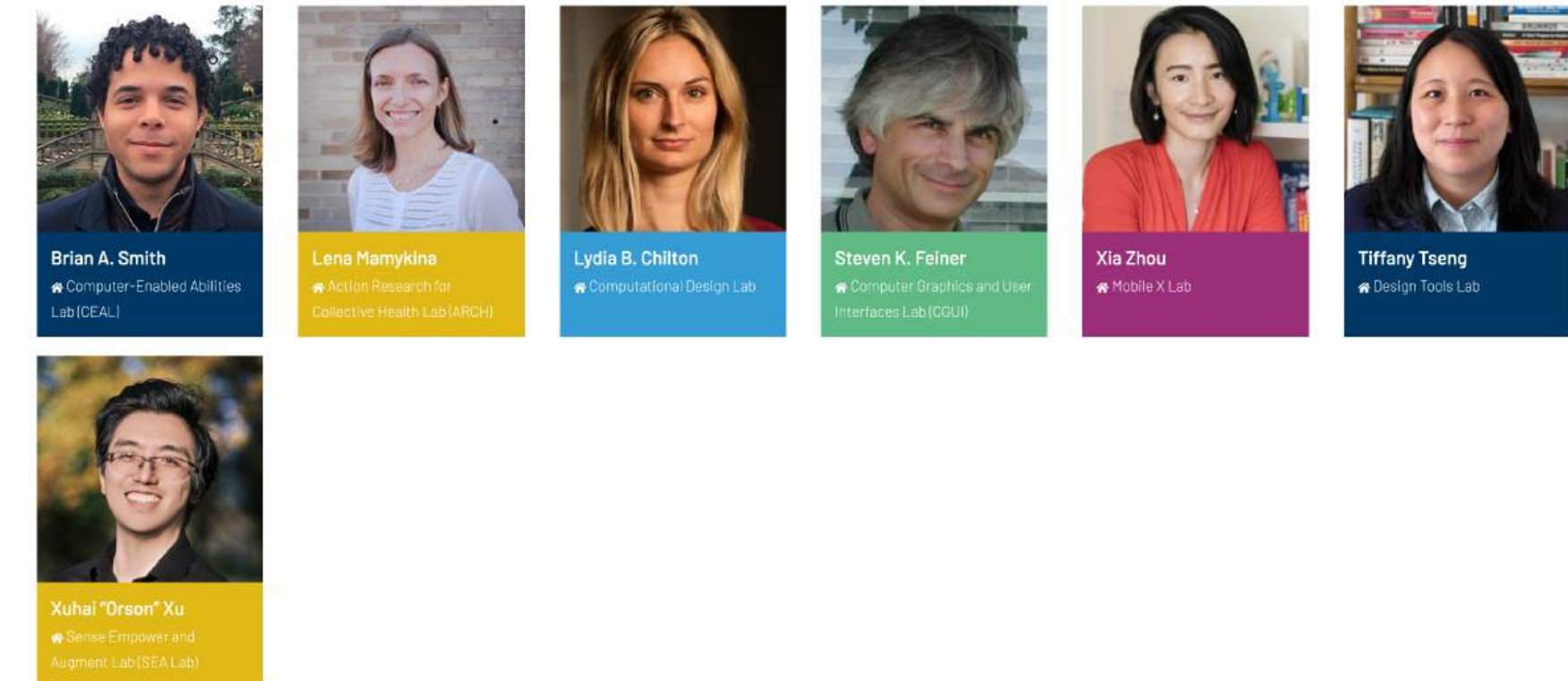
columbiahci.github.io

- UI Design (4170)
- 3D UI and Augmented Reality (4172)
- HCI Research Seminar (6178)
- Advanced Web Design Studio (6998)
- Design for GenAI / Emerging Technology (6998)
- Virtual and Augmented Reality (6998)
- Topics in Mobile X (6998)
- Methods III: Research Methods (BINF 6002)
- Systems for Human-Data Interaction (6998)
- Agentic System (6113)
- Data Visualization (4995)
- Human-Centered Design and Innovation (IEME 4200)

Human-Computer Interaction@Columbia

We're on a mission to explore how people interact with technology in various domains and exciting new ways
Check out our vibrant group of researchers who share ideas, uncover insights, and investigate the future of human-computer interaction (HCI)

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